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DEPARTMENT OF ELECTRICAL ENGINEERING & ELECTRONICS

Final report for project ‘Chemical Imaging: Robotic Scanning Platform’

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Abstract

Chemical imaging has long been a crucial area in medical and clinical research, enabling the visualization of chemical distributions across surfaces using various methods, such as Raman spectrometry, mass spectrometry, and optical spectrometry. In recent years, there is an increasing demand for remote healthcare solutions. Traditional spectrophotometers, however, remain expensive and complex to operate, posing a barrier to wider adoption.

To address these needs, this project builds upon the design principles of the open-source PySpectrometer2 to develop a compact optical spectrophotometer capable of performing linear (one-dimensional) surface scans. The system is designed with the following objectives: low cost, portability, modularity, and ease of use. Key features include a grating-based spectroscope, a Raspberry Pi camera for image acquisition, a stepper motor-powered platform, and custom-designed 3D-printed structural components modeled in SolidWorks.

Operating in the visible light range, the system incorporates a self-developed motion control and scanning algorithm, enabling synchronized actuation and real-time spectral data visualization. A dynamic weighting system was also implemented to improve data consistency and noise suppression. Preliminary results indicate strong agreement between the captured spectra and known reference spectra from calibrated light sources, demonstrating the feasibility and effectiveness of the proposed design.

Catalog

1.Introduction

Chemical imaging is a powerful technique that enables the visualization of the spatial distribution of chemical components across surfaces. It has broad applications across multiple disciplines, including biomedical sciences for tissue imaging, and clinical diagnostics. In context of emergency care, chemical imaging has been applied using spectrometers to rapidly identify chemical compositions, such as detecting early signs of drug overdose in patients arriving at Accident & Emergency (A&E) departments [1].
However, with the rapid growth of remote healthcare services, there is an increasing need to decentralize diagnostic procedures and bring them closer to patients' homes [2]. This shift demands diagnostic tools that are not only accurate but also accessible outside of traditional hospital settings.
Conventional spectrometers, while powerful, but expensive, large in scale, and require complex operating procedures. To address the challenges posed by this new usage scenario, this project aims to design and develop a spectrometer-based device that is low-cost, portable, modular, and user-friendly. The modular design will allow users or researchers to adapt the device by easily switching components to meet specific diagnostic or research requirements, making it suitable for both home-based daily use and specialized studies.
Based on the general concept design, the system can be divided into three main sections, including the spectrometer design, the linear moving platform design, and the user interface design, as shown in the following figure [3]:

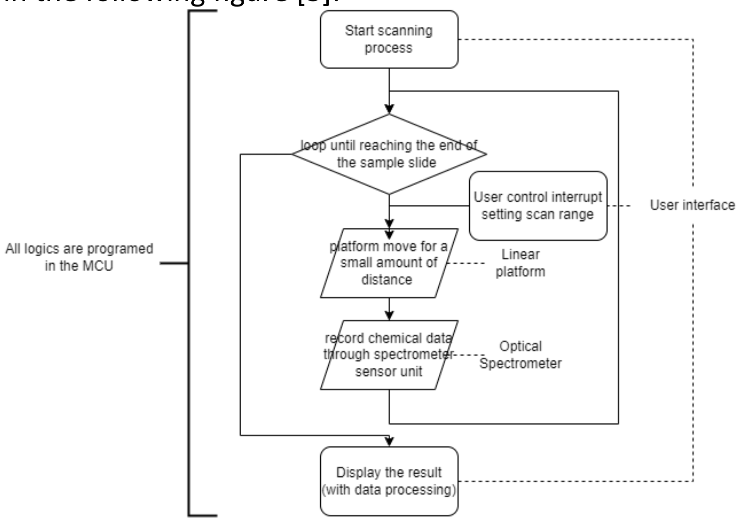


Figure 1 General concept design

This project focuses on the development of a complete functional system that integrates spectrometer data acquisition, linear platform motion control, and a user-friendly interface. This multidisciplinary approach closely aligns with the core competencies of the Mechatronic and Robotic Systems major, combining mechanical design, electronics, control theory, and software development. Designing the mechanical structure for stable and precise movement of the spectrometer platform required the application of mechanical design principles, while implementing synchronized motion and data capture involved advanced control programming and system integration. The development of the user interface further added to the challenge, demanding a clear understanding of human-computer interaction and real-time system behavior. By addressing a real-world healthcare problem, this project provided a highly practical and holistic engineering experience—strengthening my ability to analyze user needs, translate them into technical specifications.

2.Literature Survey

Chemical imaging has long been a pivotal research direction in both the medical and material sciences, offering diverse methods to inspect test substances. Throughout its history, a wide range of techniques have been developed to enhance chemical imaging. The evolution of these solutions is closely tied to the specific characteristics of the samples under investigation, leading to distinct advantages tailored to each application. This literature survey will focus on chemical imaging techniques pertinent to medical and biological research.

2.1 Optic spectroscopy & non-optic spectroscopy

In biomedical research, both optical and non-optical spectrometry techniques are employed, each offering distinct advantages. Optical spectrometry methods, including UV-Vis, fluorescence, Raman, and infrared spectroscopy, are favored for their non-invasive nature, rapid analysis, and cost-effectiveness. These techniques are extensively used in applications such as tissue imaging, metabolic profiling, and real-time diagnostics. For instance, Raman spectroscopy has been effectively utilized to detect cancer biomarkers in canine sera, demonstrating its utility in non-destructive biomedical analyses [4].
Conversely, non-optical spectrometry methods, such as mass spectrometry, offer high sensitivity and specificity, making them indispensable for detailed molecular analyses [5]. They are extensively used in proteomics, metabolomics, and lipidomics to identify and quantify biomolecules. However, these techniques often require complex instrumentation and are less amenable to portable, point-of-care applications.

2.2 Common Spectrometer Configurations

The optical configuration of a spectrometer significantly influences its performance characteristics, including resolution, sensitivity, and suitability for specific applications. Several established configurations are commonly employed in spectrometer design [6]:

2.2.1 Littrow Configuration

The Littrow configuration is one of the simplest spectrometer designs, but is not commonly seen in recent spectrometer designs, utilizing a plane reflection grating and a spherical mirror. In this setup, light enters through an entrance slit, is collimated by the mirror, and then directed onto the diffraction grating. The diffracted light is reflected along the same path, passing through an exit slit located near the entrance slit. While this configuration offers high wavelength resolution, there is a possibility of stray light and internal reflections, which can affect measurement accuracy [6].

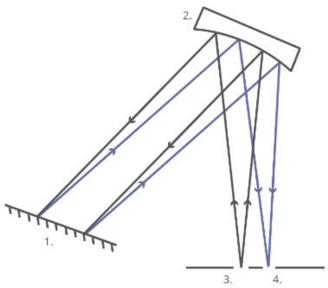


Figure 2 General illustration of Littrow configuration, 1.diffraction grating, 2.spherical mirror, 3.entrance slit, 4.Exit slit [6]

2.2.2 Ebert-Fastie Configuration

Invented by Hermann Ebert in 1889, and improved by William.G.Fastie in 1952 who improved the configuration and compensate the astigmatism problem (As shown in Figure 3). The Ebert-Fastie Configuration consists of a large spherical mirror and a plane diffraction grating, arranging the optic path as shown in Figure 4. This design reduces certain optical aberrations but may still experience issues with direct reflections and multiple diffractions [7].

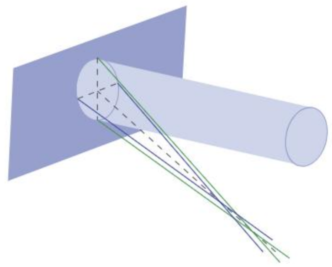


Figure 3 Astigmatism illustration, which diffraction grating causing non-overlapping focus points[6]

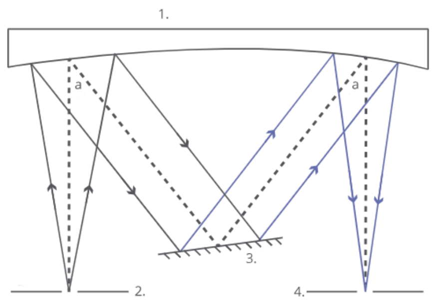


Figure 4 General illustration of Ebert-Fastie Configuration, 1.large spherical mirror, 2.intrance slit, 3. Differaction grating, 4. Exit slit [6]

2.2.3 Czerny-Turner Configuration

Iterated from Ebert-Fastie configuration. Invented by M. Czemy and A.S. Turner in 1930. The Czerny-Turner configuration is widely used in modern spectrometers due to its improved performance over earlier designs, while allowing more flexible designs. It employs two separate spherical mirrors and a plane diffraction grating. Light enters through an entrance slit, is collimated by the first mirror, diffracted by the grating, and then focused by the second mirror onto the detector through the exit slit. This arrangement minimizes stray light and optical aberrations, providing better spectral resolution and accuracy [8].

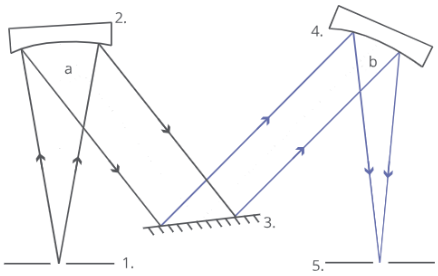


Figure 5 General illustration of the Caemy-Turner configuration, 1. Entry slit,2. First mirror,3. Diffraction grating,4. Second mirror,5. Exit slit

2.2.4 Concave Aberration-Corrected Holographic Grating Configuration

In this configuration, a concave holographic diffraction grating serves both to disperse and focus the light, eliminating the need for additional mirrors. Light enters through an entrance slit, is diffracted and focused by the grating, and then exits through an exit slit onto the detector. This design simplifies the optical path and reduces aberrations, making it suitable for compact and portable spectrometers.

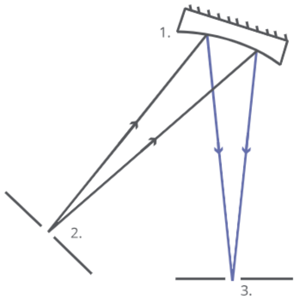


Figure 6 Illustration of the Concave Aberration-Corrected Holographic Grating Configuration, 1. Concave holographic, diffraction grating,2. Entry slit,3. Exit slit

2.3 Miniaturization of Optical Spectrometers

Since the early 1990s, extensive development efforts have been devoted to the miniaturization of optical spectrometers, resulting in multiple different types of designs that span different spectral ranges and resolutions. Mentioned in Yang's detailed literature review, these efforts have led to the classification of miniaturized spectrometers into four principal categories, each defined by distinct working principles and technological trade-offs [9].

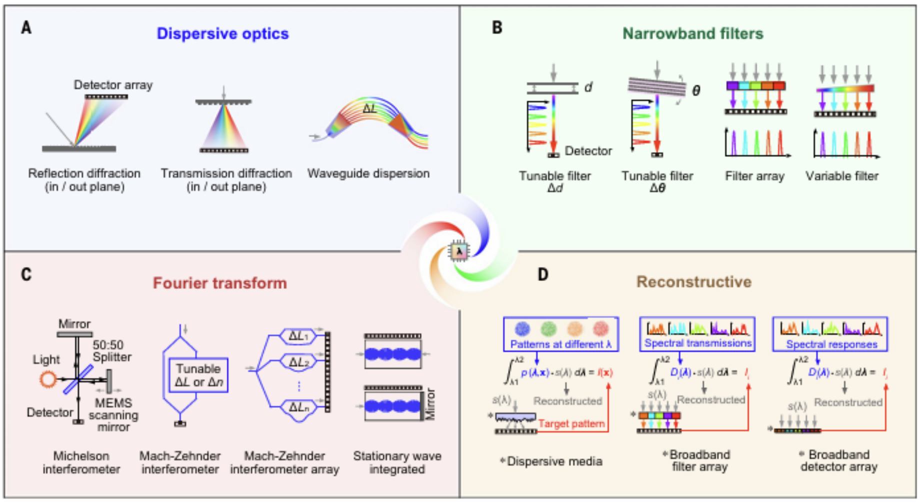


Figure 7 Different development direction for miniaturized spectrometer systems in the past 30 years, A. Miniaturized dispersive optics, B. Narrowband filters, C. Fourier transform-based, D. Computational spectral reconstruction [9]

The earliest miniaturized systems relied on dispersive optics, using reflection or transmission gratings and detector arrays to spatially resolve incident light by wavelength (Figure 7A). Parallel to this, narrowband filter-based spectrometers (Figure 7B) emerged, which selectively transmit certain spectral components via fixed or tunable filters. These designs, although compact, often rely on mechanical tuning or large filter arrays, limiting their integration potential.

A third major approach is the Fourier transform (FT) microspectrometer (Figure 7C), which uses interferometric principles to extract spectral information from temporal or spatial interference patterns. Implementations based on Michelson or Mach-Zehnder interferometers, often incorporating MEMS scanning mirrors, offered high resolution in a small form factor but still required precise alignment and relatively complex fabrication.

As the field evolved, these three categories—dispersive, filter-based, and FT spectrometers—shifted toward more planar and integrated architectures, such as waveguide-based photonic circuits, enabling further size reduction and improved manufacturability. Most recently, a fourth paradigm has gained prominence: the reconstructive (or computational) spectrometer (Figure 7D). These devices leverage the increasing availability of high-performance, low-cost computing to reconstruct incident spectra from complex optical response patterns, using techniques such as compressive sensing and machine learning. Instead of relying on precise spatial dispersion or filter tuning, reconstructive systems encode spectral information into multiplexed responses across broadband filters or detectors and then computationally infer the spectrum.

This progressive evolution—from physically resolving spectrometers to computationally driven designs—reflects a broader trend in optics: shifting complexity from hardware to software. Such strategies are especially valuable in applications where miniaturization, power efficiency, and integration outweigh the need for ultra-high resolution [9].

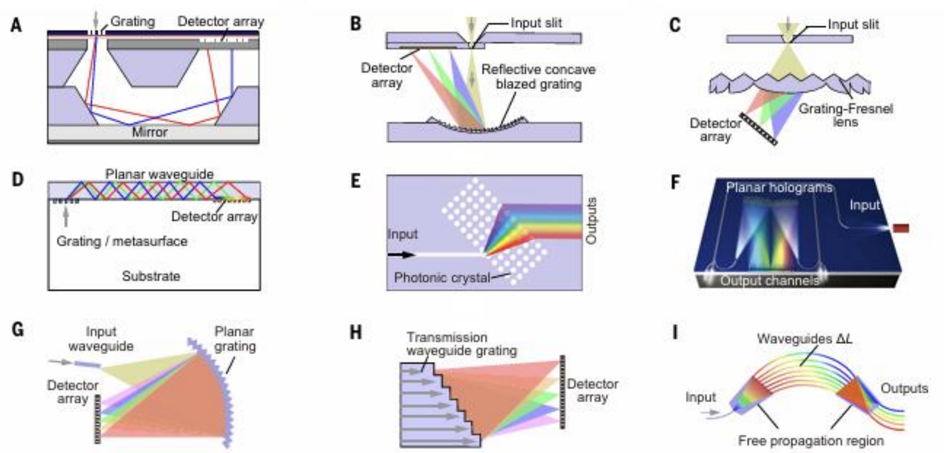


Figure 8 Different mechanical configurations of spatially dispersive spectrophotometers, A&B. Miniaturized spectrometer systems based on out-of-plane spatial dispersion using planar and concave gratings, respectively, C. grating-Fresnel spectrometer, D. buried grating on a waveguide sensor, E. photonic crystal–based grating, F. holographic element, G. planar echelle grating, H. transmission waveguide grating, I. arrayed waveguide grating [9]

2.4 Practice of compact spectrophotometers

In recent years, there have been some innovations in the design of compact, low-cost, and customizable spectrophotometer systems. These advancements are motivated by the need for portable and accessible spectral analysis tools across scientific, educational, and field applications. However, none of the designs are able to perform linear scanning capability.

2.4.1 Open-Source Miniature Spectrophotometer (OSMS)

Laganovska et al. introduced a low-cost, portable open-source spectrophotometer based on the Hamamatsu C12880MA sensor chip. The system, housed in a 90×85×58 mm 3D-printed case, uses an Arduino Nano with Bluetooth and USB connectivity. It supports absorbance measurements from 450–750 nm, with a 15 nm resolution, and allows real-time data acquisition via an Android app or PC interface. The design emphasizes ease of replication and customization. Validation studies showed performance comparable to moderate-cost commercial devices in Vitamin B12, phosphate, and enzyme activity measurements [10].

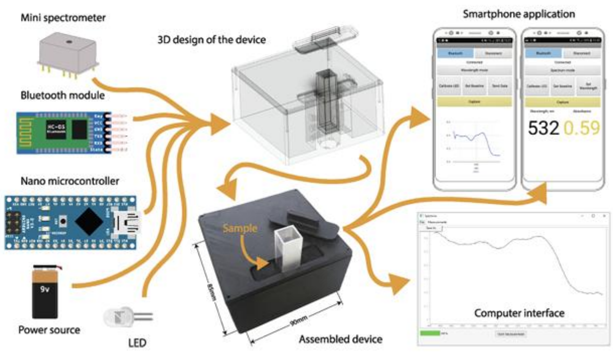


Figure 9 The main components of the device, (the mini spectrometer is C12880MA chip) [10]

2.4.2 Optical Fiber-Based Spectrometer System

An enhanced spectrometer system developed by Tunens et al. utilizes a fiber-based design controlled by a Raspberry Pi. Unlike previous designs limited to transmission mode, this version supports fluorescence, absorption, and light scattering measurements through interchangeable fiber and LED modules. Operating in the 400–700 nm range, the system uses a high-precision 16-bit ADC and supports extensive customization, while also using the C12880MA compact spectrometer chip. Calibration against commercial systems (e.g., FLS1000) showed strong fidelity for luminescence and absorption spectra, highlighting the platform's versatility for scientific applications [11].

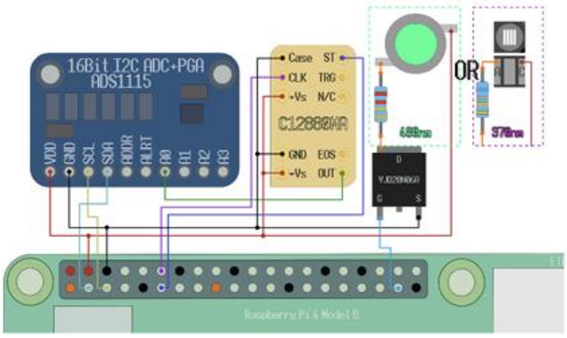


Figure 10 General circuit design of the optic Fiber-based spectrometer [11]

2.4.3 On-Chip Stratified Waveguide Filter (SWF) Spectrometer

Li and Fainman presented a cutting-edge miniaturized spectrometer based on 32 stratified waveguide filters integrated onto a silicon chip. This design enables single-shot spectrum reconstruction with a total footprint of just 35×260 µm². It achieves ~0.45 nm spectral resolution across a 180 nm bandwidth centered at 1550 nm. The approach demonstrates remarkable potential for ultra-compact, high-resolution, CMOS-compatible spectral sensing in portable electronics and IoT platforms [12].

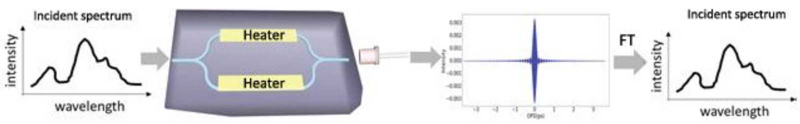


Figure 11 Illustration of general structure of Fourier Transform Spectrometer used in this project [12]

Figure 11 illustrates a Fourier Transform Spectrometer (FTS) on the silicon photonics platform, where the incident light passes through a balanced Mach–Zehnder Interferometer (MZI) with integrated heaters that vary the optical path delay (OPD) between its two arms. This modulation produces an interferogram—essentially the autocorrelation of the input signal—which is then mathematically transformed using a Fourier Transform (FT) to reconstruct the original optical spectrum. This approach enables compact, high-resolution spectral analysis suitable for on-chip integration [12].

2.4.4 Theremino Spectrometer with Linear Sensor

The Theremino project offers a software solution allowing linear sensor input and camera input. Allowing dynamic spectrometer designs, and offers a DIY-friendly linear-sensor-based spectrometer design demo emphasizing accessibility and practicality. Constructed primarily from beech wood and 3D-printed or repurposed materials (e.g., shampoo bottles for diffusers), it supports user-adjustable components like slits and diaphragms to optimize spectral clarity. While resolution and sensitivity depend on construction quality and sensor choice [13].

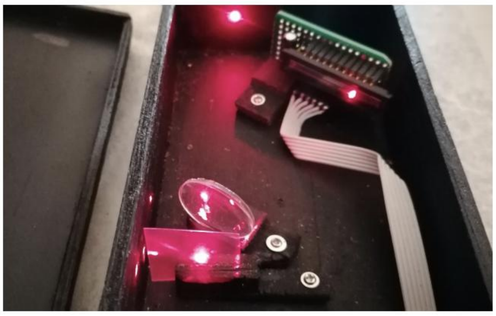


Figure 12 General structure of the linear sensor-based spectrometer [13]

2.4.5 PySpectrometer2, Python-Based DIY Spectrometer Platform

The PySpectrometer2 project by Les Wright is a flexible, open-source spectrometer toolkit written in Python and designed for educational and hobbyist use. It supports a wide variety of USB-connected spectrometer hardware (e.g., Hamamatsu, Ocean Optics) and includes a real-time graphical user interface for spectrum display, peak analysis, and wavelength calibration. The system also supports camera-based spectrometers using OpenCV for low-cost DIY builds. Designed with modularity in mind, it can be extended with custom spectral processing routines or integrated into other Python-based scientific workflows [14].

3. Industrial relevance, real-world applicability/societal impact

The development of a robotic scanning platform for chemical imaging represents a convergence of robotics, biomedical engineering, and analytical spectroscopy with significant implications across both healthcare and industrial domains. Designed specifically to support early detection of drug overdoses in pediatric patients, this project addresses a pressing healthcare challenge while offering scalable technological innovation with broader utility across sectors.

3.1 Market Trends and the growing need for Portable Spectrometers

The global portable spectrometer market is experiencing significant growth, driven by the increasing demand for compact, affordable, and field-deployable analytical tools. Valued at approximately USD 2.16 billion in 2024, the market is projected to reach USD 3.9 billion by 2031, growing at a compound annual growth rate (CAGR) of 8.5% [15]. This trend is attributed to the expanding demand of portable spectrometers across various sectors, including healthcare, environmental monitoring, and industrial quality control.
In remote healthcare settings, the need for portable spectrometers is particularly pronounced. Devices like the Blaise™ portable Raman spectrometer exemplify this trend, offering compact designs that integrate with smartphones via USB-C, enabling bio-surveillance and field deployments without bulky hardware [16].

The COVID-19 pandemic has further underscored the importance of accessible medical equipment at home. The widespread adoption of at-home COVID-19 testing kits has shifted public expectations, leading to increased demand for home-use medical devices [17]. This trend has extended beyond infectious disease testing, with consumers seeking convenient solutions for monitoring various health parameters, thereby driving the growth of the home medical equipment market.

3.2 Modular Design: Versatility Across Diverse Applications

he modular nature of the spectrometer developed in this project allows for adaptability to a wide range of applications:

- **Emergency Medicine and Overdose Scenarios**: In the context of emergency care, particularly in Accident & Emergency (A&E) departments, time is often the most critical factor. Rapid identification of ingested or injected substances can drastically influence treatment protocols and patient outcomes. A modular spectrometer system can be calibrated and pre-configured with libraries of spectral fingerprints associated with common prescription and illicit drugs. This allows healthcare professionals to perform immediate, on-site chemical analysis of biological samples such as vomit, saliva, or urine, reducing reliance on time-consuming lab-based methods.
  Because the device is portable and user-friendly, it can also be operated by paramedics or emergency responders, effectively bringing hospital-grade analytical capacity directly to the field. By swapping in drug-specific filters, modifying illumination sources, or updating software libraries, the same base device can adapt to evolving drug trends and localized needs, which is particularly important in managing regional overdose crises.
- **Biological Research:** Modularity greatly benefits biological research, where a single tool often needs to support multiple investigative functions. The spectrometer system can be adjusted for absorbance, fluorescence, and reflectance modes by reconfiguring optical paths and detectors. This means the same core device can be used to study protein concentrations, cell growth rates, or chemical markers without the need for multiple instruments.
- Furthermore, research-grade experiments frequently demand customization—different wavelengths, exposure timings, or sample holders—which can be easily accommodated with a modular platform. It becomes a flexible research assistant rather than a single-use tool. Academic labs with limited funding especially benefit from such systems, as the spectrometer can grow with the scope of the research by simply adding or replacing modules rather than purchasing entirely new equipment [18].
- **Remote Healthcare:** The modular spectrometer can also play a transformative role in remote healthcare delivery, especially for populations living in rural or underserved regions. Its lightweight, portable, and intuitive design enables local caregivers or even patients themselves to operate the device with minimal training. Affordability make the device a valuable tool for improving medical diagnostics in low-income regions, where access to advanced laboratory equipment is limited [16].
- **Education and Personal Modification:** Another valuable use of a modular spectrometer system is in STEM education and personal innovation. 3D-printable housings, open-source hardware designs, and simple GUI interfaces allow students

and educators to assemble, calibrate, and operate the device as part of their coursework. This helps build not only an understanding of spectroscopy but also skills in optics, electronics, and data analysis.

Because each component—from diffraction gratings to light sources—can be individually replaced or upgraded.

3.3 Integration with Robotic Platforms: Enhancing Capabilities

The spectrometer's modular design also allows for integration with robotic platforms, expanding its utility Combining the spectrometer with robotic systems, such as quadrupedal robots, can enhance environmental sensing capabilities. For instance, integrating spectroscopic methods with electronic noses enables the detection of hazardous substances in dangerous environments, facilitating applications in industrial safety and disaster response. [19]

3.4 Scientific and Societal Impact

Scientifically, the convergence of robotic platform, and spectroscopy is enabling more efficient real-time, high-resolution chemical imaging. This multidisciplinary approach facilitates the development of systems capable of rapid substance identification, which is crucial in emergency medical scenarios.

Societally, the deployment of such integrated diagnostic platforms has the potential to significantly improve patient outcomes by enabling early detection of drug overdoses. This is particularly impactful in pediatric care, where swift diagnosis is essential for effective treatment. Moreover, the scalability and adaptability of these technologies can contribute to broader public health initiatives, including community-based overdose prevention programs and environmental monitoring for hazardous substances.

The development of this robotic platforms integrated with advanced chemical imaging technologies holds substantial promise for enhancing diagnostic capabilities across various sectors.

4.Theory

As outlined in the introduction, this project comprises three major components: the Spectrophotometer design, the linear moving platform design, and the user-interface development. Accordingly, the Theory section is structured to address the relevant theoretical foundations underlying each of these sectors.

4.1 Spectrophotometer Section

4.1.1 Thin Layer Chromatography (TLC)

Thin Layer Chromatography (TLC) is a widely used analytical technique for separating and identifying compounds within a mixture. It operates on the principle of differential adsorption, where components of a mixture migrate at different rates over a stationary phase under the influence of a mobile phase. As the solvent ascends by capillary action, compounds distribute based on their affinities, resulting in separation along the plate[20].

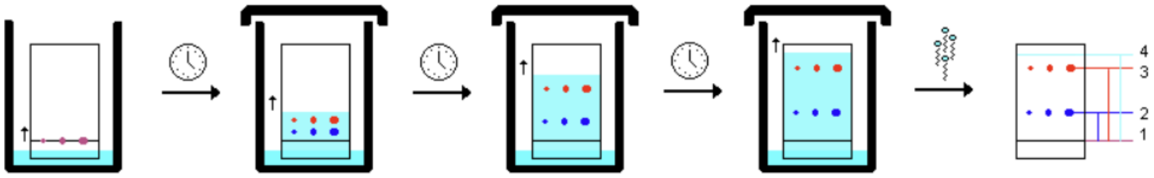


Figure 13 General illustration of TLC process, 1-2:the distance travelled by compound A, 1-3 the distance travelled by compound B, 1-4: distance travelled by solvent [20]

After development, the separated compounds are typically visualized using methods such as ultraviolet (UV) light exposure or chemical staining. The position of each compound is characterized by its retention factor (Rf), calculated as [20]:

$$R_f = \frac{\text{distance traveled by sample}}{\text{distance traveled by solvent}}$$

Figure 14 The formula used to calculate the retention factor.

Rf values are specific to individual compounds under consistent experimental conditions and can be used for identification by comparison with known standards. For quantitative analysis, techniques like densitometry can measure the intensity of compound spots, allowing for concentration estimation. However, traditional methods may lack precision and are time-consuming, especially for complex mixtures. Integrating TLC with digital imaging and spectroscopic techniques enhances accuracy and efficiency, facilitating more precise identification and quantification of compounds.

4.1.2 The Beer-Lambert Law

The Beer-Lambert Law, also known as Beer's Law, describes the linear relationship between the absorbance of light by a substance and its concentration in solution.
Mathematically, the Beer-Lambert Law is expressed as:

$$A = \varepsilon c l$$

A	Absorbance	
ε	Molar absorption coefficient	$M^{-1}cm^{-1}$
c	Molar concentration	M
l	optical path length	cm

Absorbance A can also be expressed by the incident light intensity I0 and the transmitted light intensity I by the following equation:

$$A = \log_{10}\left(\frac{I_o}{I}\right)$$

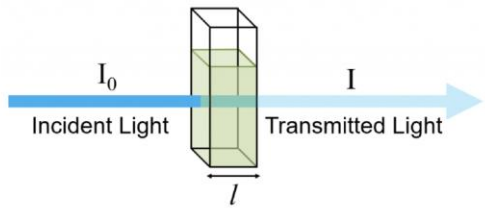


Figure 15 Indication of I_0 and I [21]

4.1.3 Diffraction Grating

A diffraction grating is an optical component with a periodic structure that diffracts incident light into several beams traveling in different directions. The directions of these diffracted beams depend on the spacing of the grating and the wavelength of the light.

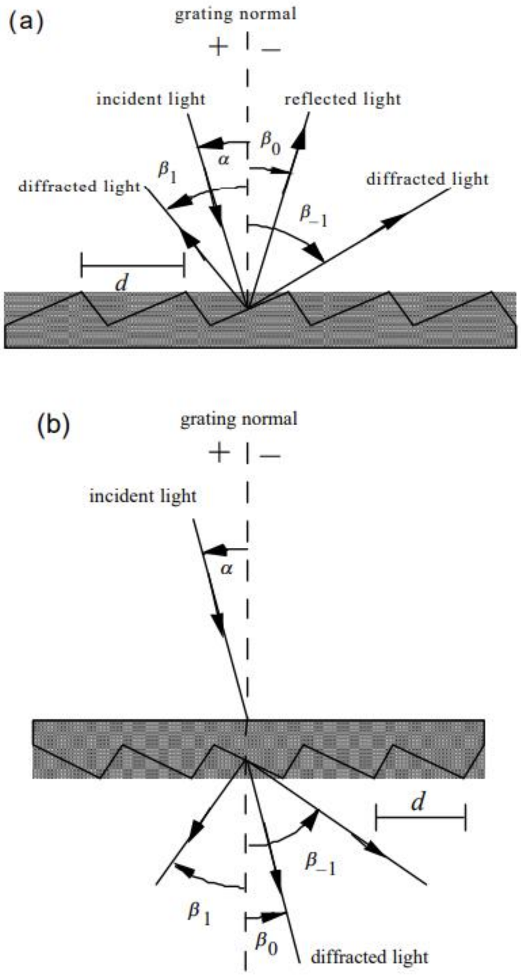


Figure 16 Illustration of working principle of diffraction grating, (a) reflection grating, (b) transmission grating , α is referring to the incident angle, d is the groove spacing, β_0 is the 0 order diffraction, β_1 and β_{-1} are the angle of positive and negative first order diffraction[22]

As shown in the figure, the diffraction grating can be used to perform reflection grating and transmission grating. While the reflection grating is more commonly seen in spectrometer design, as mentioned in the 2.2 section, where all the general configurations are using reflection grating. However, the grating spectroscope, on the other hand, is an example of a transmission grating.

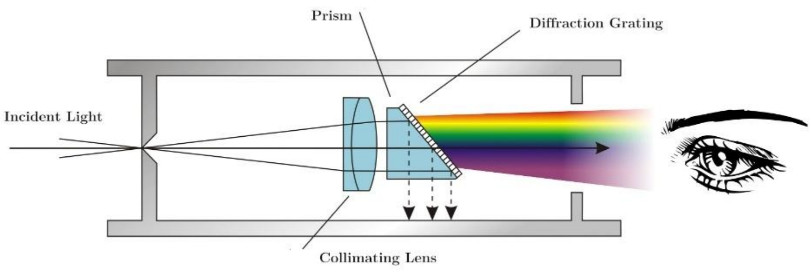


Figure 17 General illustration of grating spectrometer configuration [23]

As shown in Figure 16 The relation between the incident light and diffracted light can be concluded as following diffraction grating equation:

$$m\lambda = d\,(\sin\alpha + \sin\beta)\ (m = \pm1, \pm2, \pm3, \ldots)$$

Where **d** is the groove spacing, **α** is the angle of incident light, **β** is the angle of diffraction light, **λ** is the wavelength of the light, and **m** is an integer that refer to the order of diffraction. To understand m, each order corresponds to a condition where the path difference between rays diffracted from adjacent grooves is an integer multiple of the wavelength. This is required for constructive interference. Based on the equation it can be also obtained that $-2d<m\lambda<2d$ so that the resulting diffraction angle is physically meaningful [22].

Based on the diffraction grating equation, it can also be concluded that the dispersion produced by a diffraction grating is not linear across all wavelengths. The angular separation between wavelengths increases with wavelength, leading to a non-uniform spread of the spectrum. This characteristic must be considered when designing optical systems that incorporate diffraction gratings, especially when precise wavelength measurements are required.

4.1.4 Pi Camera and OV5647 Sensor: Light-to-Digital Conversion

The Raspberry Pi Camera Module utilizes the OmniVision OV5647 CMOS image sensor to capture visual information. This process involves several stages, transforming incoming light into a digital image represented in the RGB8888 format.

At the core of the OV5647 sensor are millions of photodiodes arranged in a grid, each corresponding to a pixel. When photons strike these photodiodes, they generate electron-hole pairs through the photoelectric effect. The accumulation of these charge carriers results in an electrical signal proportional to the light intensity. This process effectively converts the optical image into an array of analog electrical signals.

To discern color, the sensor employs a Bayer filter mosaic over the photodiode array as shown in the following figure, allowing each photodiode to capture light intensity for a specific color channel. Typically, the pattern includes twice as many green filters as red or blue to mimic human visual sensitivity to green light [24].

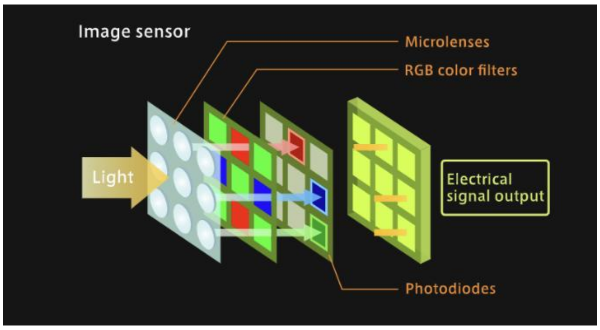


Figure 18 General illustration for RGB filter structure of the image sensor[24]

Each pixel's analog signal is first amplified to enhance the signal-to-noise ratio. Subsequently, these signals are converted into digital values using on-chip analog-to-digital converters (ADCs). This conversion translates the continuous analog signals into discrete digital numbers, representing the intensity of the respective color channels. The final image is formatted into the RGB8888 format, where each pixel is represented by 32 bits. This format allows for a wide range of colors and is compatible with various image processing applications.

In many analytical applications, it is a common procedure to convert these color images into grayscale. Grayscale images simplify the data by reducing each pixel to a single intensity value. The conversion from RGB to grayscale isn't a mere averaging of the three channels; instead, it employs a weighted sum to account for the human eye's varying sensitivity to different colors. This perceptual bias is quantitatively represented in the standard luminance formula

$$\text{Grayscale Intensity}=0.2126\times R+0.7252\times G+0.0722\times B$$

This weighted approach ensures that the resulting grayscale image closely aligns with human brightness perception, preserving the visual nuances of the original scene [25].

4.2 Linear moving platform section

4.2.1 Screw rod mechanism

In the context of the linear moving platform design, the screw rod mechanism plays a pivotal role in converting rotational motion into precise linear displacement. Two primary types of screw mechanisms are commonly employed: lead screws and ball screws.

A lead screw consists of a threaded shaft and a matching nut as shown in Figure 18. As the shaft rotates, the nut moves linearly along its threads. This mechanism relies on sliding contact between the threads of the screw and the nut, which results in higher friction and lower efficiency compared to rolling mechanisms. However, this friction provides a self-locking feature, preventing the screw from back-driving under load, making lead screws suitable for vertical applications where holding position without power is essential [26].

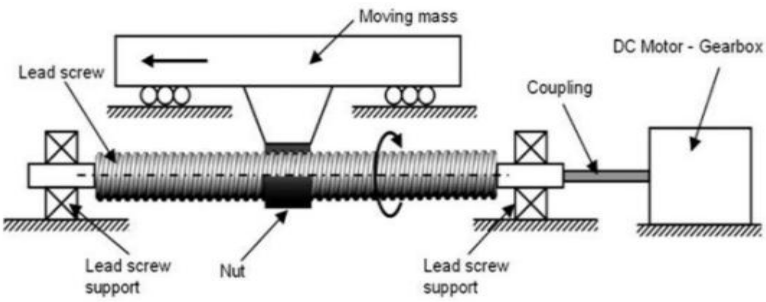


Figure 19 General illustration of lead screw configuration [26]

Ball screws incorporate ball bearings between the screw shaft and the nut as shown in Figure 19, facilitating rolling contact instead of sliding. This design significantly reduces friction, resulting in higher efficiency and precision.

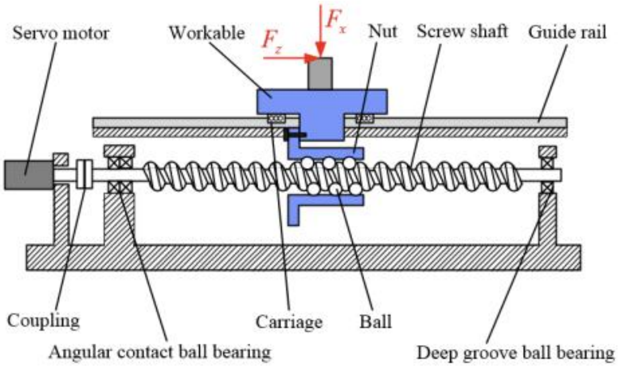


Figure 20 General illustration of Ball screw configuration [27]

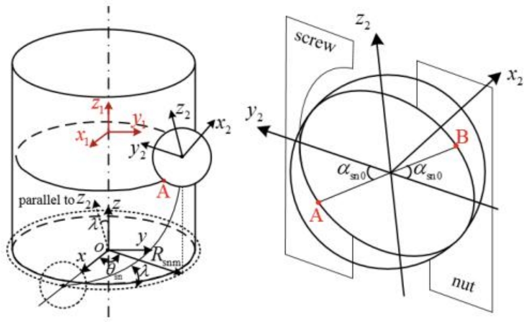


Figure 21 The relationship between coordinate system in calculation of Restoring Force of ball screw, [27]

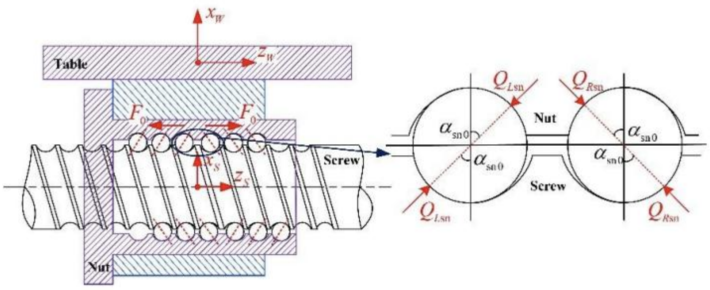


Figure 22 Detail Force diagram of ball screw system under payload, Q_{Lsn} and Q_{Rsn} are the Restoring force generated at steady state [27]

Based on the research by C. Liu, C. Zhao, Z. Liu, and S. Wang on the dynamic behaviour of ball screw feed systems [27]. Compared to conventional lead screw systems, ball screw configurations offer superior dynamic performance, primarily due to their ability to generate higher restoring forces with significantly lower friction. The restoring force in a ball screw arises from the elastic deformation at the contact interface between the recirculating balls and raceways, enhanced by the application of preload. This elastic response enables the system to resist external disturbances with greater stiffness and recover more precisely to its original position, which is critical for applications requiring high positional accuracy and stability under dynamic loading. In contrast, lead screws rely on sliding contact, which not only results in lower efficiency but also produces less predictable and often lower restoring forces due to greater frictional losses. While lead screws can be self-locking, their increased internal friction reduces responsiveness and precision in high-speed or high-frequency applications. Thus, the ball screw's ability to generate strong, controllable restoring forces makes it an outstanding choice for precision machinery, CNC systems, and robotic actuators, where high efficiency and dynamic stiffness are prioritized over self-locking behaviour.

4.2.2 Stepper motor control

Unlike conventional motors that rotate continuously, stepper motors move in discrete steps, allowing for accurate control of position and speed without the need for feedback systems.

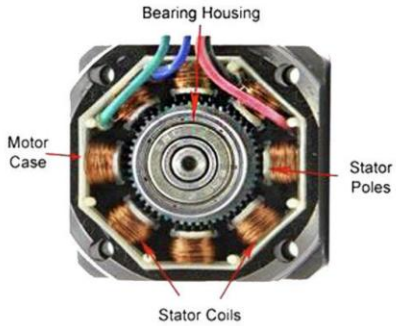


Figure 23 Inner structure of Stepper motor[28]

The general structure of stepper motor is simple, as shown in the Figure 23, including the stator and the rotor. When electrical pulses are supplied to the stator windings in a specific sequence, they create a rotating magnetic field. This field interacts with the rotor, causing it to align with the magnetic field and move in fixed angular increments, known as steps.
Commonly, Stepper motors are typically controlled using an open-loop system, where the position is determined by the number of input pulses rather than feedback from sensors. Each pulse advances the motor by one step, and the speed of rotation is proportional to the frequency of these pulses. This method simplifies the control system and reduces costs, making stepper motors ideal for applications requiring precise positioning without complex feedback mechanisms [29].

4.3 User interface section

4.3.1 Multithreading

Multithreading enables embedded systems to handle multiple tasks simultaneously, improving efficiency and real-time performance. Each thread manages specific functions like sensor monitoring, motor control, or communication. Typically coordinated by a Real-Time Operating System (RTOS), multithreading ensures task prioritization and timing accuracy.

However, it introduces challenges like shared resource management, requiring synchronization tools such as mutexes and semaphores[30].

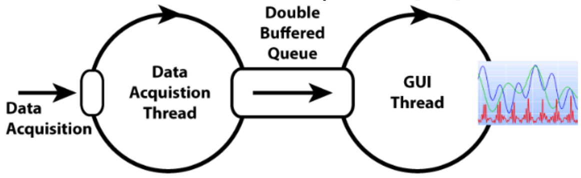


Figure 24 Illustration of Multithreading for real time data plotting [31]

5. Design

As this project focuses on system development, it integrates both hardware and software components to achieve its objectives. Therefore, this section provides a comprehensive overview of the design process, beginning with an illustration of the overall system architecture, then details the hardware design, followed by the software development, highlighting how each component contributes to the functionality and performance of the system.

5.1 General system design

Based on the general requirement of the project, as mentioned in the 1. Introduction section, the system can be divided into three primary sections, including the spectrometer, the linear moving platform, and the user-interface.

The initial concept for the spectrometer component involved replicating the OpenRAMAN Starter Edition, an open-source Raman spectrometer designed for educational purposes. This design offers a theoretical spectral resolution ranging from 0.35 nm at the sample centre to 1.3 nm at the edges, which is promising for various applications [32].

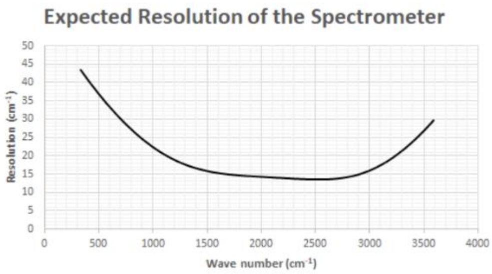


Figure 25 Expected resolution of the Raman spectrometer [32]

However, upon deeper investigation, several limitations became apparent. Raman spectroscopy requires a high-power laser, typically classified as Class 3B, to excite the sample and generate the Raman spectra. Class 3B lasers pose significant safety risks, this also violate the safety restriction of the Final Year Project Safety Guidelines.

Figure 26 Illustration of Raman spectrometer system [32]

Moreover, miniaturizing the Raman spectrometer introduces additional challenges. It requires complex recalculations of the optical path and often requires custom-designed lenses, leading to increased development time and costs for each iteration. These factors make the Raman spectrometer design misaligned with the project's objectives, especially concerning budget constraints and timeline management.

Due to the limitation in laser usage, the new version concept design was focus on visible light span spectrometry, generally based on the PySpectrometer2 open project [14], which is combining a grating spectroscope and capture the spectrum footage using a Raspberry Pi camera.

Figure 27 [14]

Given the requirement for the linear moving platform to deliver high positional accuracy and repeatable motion, while not necessitating high-speed operation, a stepper motor was identified as the most suitable actuation method. As mentioned in the 4.2.2 section, stepper motors offer precise control over movement through discrete steps, making them ideal for applications where exact positioning is critical, simplifying system development. The ability to hold position without feedback in open-loop configurations also makes stepper motors a cost-effective solution compared to servo systems, aligning well with the project’s performance and budget constraints. Taking inspiration from DIY 3D printers, the linear platform would also be equipped with End stop for safety and position initialization.

With the core concepts for the spectrometer and the linear moving platform established, the selection of a suitable microcontroller unit (MCU) became clear with specific demands. Several widely used MCUs were evaluated for this project, including the Arduino Uno, STM32F103C8T6, ESP32, Raspberry Pi 4 Model B, and Raspberry Pi 5. Each offers varying levels of processing capability, I/O configurations, and software support.

The following table summarizes the key specifications of the considered MCUs:

Chart 1 Common MCUs specifications

MCU	**Processor**	**I/O Capabilities**	**Price**	**Programming Language Support**

Arduino Uno	8-bit ATmega328P @ 16MHz	14 digital I/O, 6 analog inputs	£32	C/C++ (Arduino IDE)
STM32F103C8T6	32-bit ARM Cortex-M3 @ 72MHz	37 GPIOs, USB, SPI, I2C, UART	£8	C/C++ (STM32CubeIDE)
ESP32	Dual-core Xtensa LX6 @ 240MHz	Wi-Fi, Bluetooth, multiple GPIOs	£14	C/C++, MicroPython, JavaScript
Raspberry Pi 4 Model B	Quad-core Cortex-A72 @ 1.5GHz	40 GPIOs, USB, HDMI, CSI, Ethernet	£71 (8G RAM)	Python, C/C++, Java
Raspberry Pi 5	Quad-core Cortex-A76 @ 2.4GHz	Enhanced I/O, PCIe, dual 4K HDMI	£87.5(8G RAM)	Python, C/C++, Java, Scratch

The Raspberry Pi series clearly stands out in terms of computational power and flexibility. The Raspberry Pi 4 Model B provides a robust platform capable of simultaneously supporting HDMI OS desktop output, precise control of the linear platform, and real-time camera data acquisition through its onboard CSI interface. Although the Raspberry Pi 5 offers even higher performance, it was ultimately not selected due to its relative novelty and current lack of mature software and library support, which could complicate development and integration. Thus, the Raspberry Pi 4 Model B was chosen as the most balanced and reliable option for this application.

To summarize, the system is composed of three integrated subsystems: the optical spectrometer, the linear motion platform, and the graphical user interface. At the core of the system is the Raspberry Pi 4B, which coordinates data acquisition, motion control, and user interaction. The optical path begins with a parallel LED light source illuminating the sample, with the transmitted or reflected light captured by a CSI camera. This camera is directly connected to the Raspberry Pi and relays footage for spectrum extraction. The platform's motion is driven by a 42-stepper motor, controlled via a TB6600 motor driver, which receives pulse signals from the Raspberry Pi. An end-stop switch provides positional feedback to prevent overtravel. Finally, a 10.1-inch monitor displays the GUI, enabling the user to configure parameters, view live data, and monitor scanning progress.

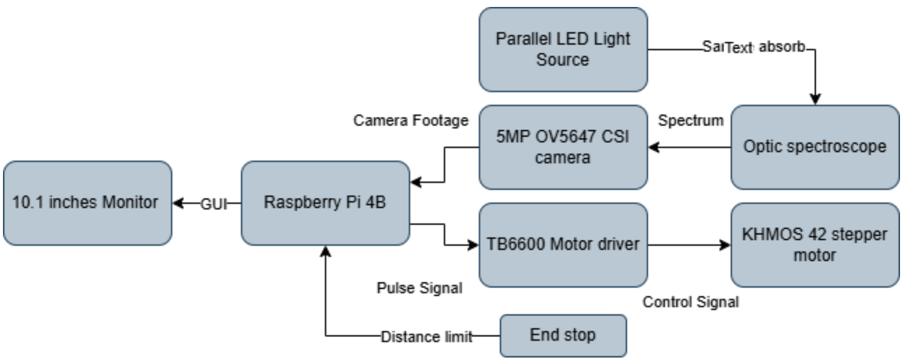


Figure 28 Illustration of general system design

5.2 Hardware design

Guided by the conceptual framework established in the previous section, the hardware development process was organized into three main phases: linear platform design, spectrometer integration, and the overall structural assembly. Each of these will be discussed in detail in the following subsections.
To efficiently integrate multiple components into a cohesive system within a limited timeframe—and to support rapid iteration at minimal cost—the primary fabrication method for custom-designed parts was Fused Deposition Modeling (FDM) 3D printing, using PLA+ as the chosen material.

5.2.1 Linear moving platform design

As shown in Figure 29 the core linear platform design, which is directly adapted from industrial 3D CNC systems. The main structural component is an aluminum profile, offering both mechanical strength and a modular base for mounting. Fiberglass plates are secured on both ends to enhance overall stability.
A stepper motor with screw as shown in Figure 30 is mounted on the Fiberglass plate at one side of the aluminum profile and is coupled to a ball screw, as previously discussed in Section 4.2.1. The ball screw was selected over a conventional lead screw due to its superior precision and efficiency, making it better suited for applications requiring accurate linear motion. The screw has a lead of 4 mm, meaning the platform move 4 mm for every full rotation of the screw, assume running the stepper motor at 3200 microstep per revolution, then he scanning resolution could reach 0.00125mm per step.

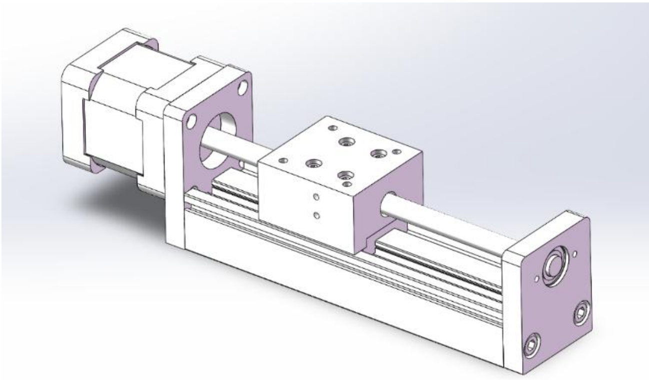


Figure 29 General design of the core structure of linear moving platform

Figure 30 Stepper motor with screw mounted

The opposite end of the screw is supported by a bearing to maintain alignment and minimize friction during operation. Linear guide rods run parallel to the screw along the aluminum profile, serving as precise tracks for the sliding block.
The moving platform is mounted onto the sliding block and interfaces with the ball screw via a ball nut, allowing synchronized and smooth linear motion. The designed travel distance is 100 mm, which is sufficient to accommodate two linear samples side by side. This length also aligns with standard rail dimensions available from online suppliers, avoiding the need for custom fabrication.
Importantly, this assembly is a direct adaptation of a standard industrial linear module. All key components—including the stepper motor, ball screw, linear guides are readily available from online marketplaces. This approach minimizes cost, simplifies assembly, and eliminates the need for complex modifications, making it ideal for rapid, low-budget prototyping.

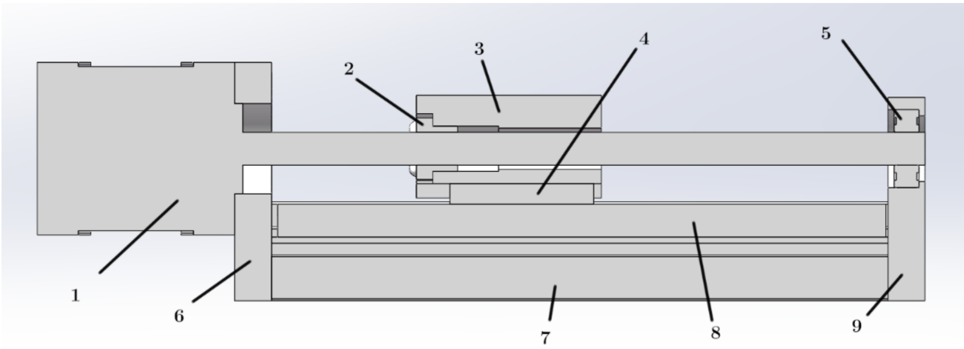


Figure 31 Cross-section view of the platform core, 1. Stepper motor with screw, 2. Ball bearing, 3. Platform, 4. Guide rail slider, 5. Bearing, 6 and 9. Fiberglass plates, 7. Aluminum profile, 8. Guide rail

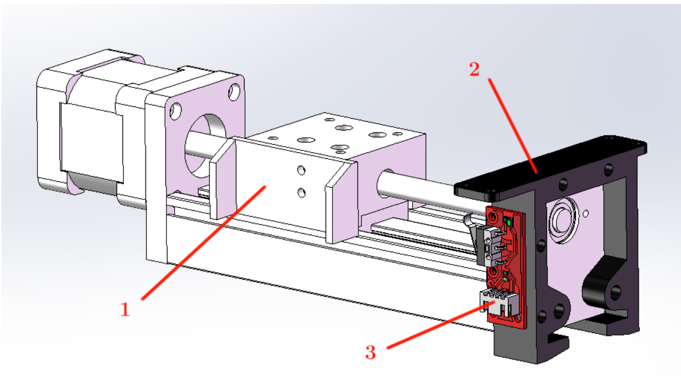


Figure 32 Design of Endstop system, 1. Endstop trigger mechanism mounted on the platform, 2. 3D printed structure for mounting Endstop and further structures, 3. Endstop

To implement end-position detection, I designed and 3D printed custom mounting parts, as shown in Figure 32, to hold an endstop switch securely in place. These parts also include a small trigger mounted on the moving platform, enabling reliable and repeatable activation of the endstop during operation.

5.2.2 Spectrometer Design

According to the general concept design, the spectrometer system consists of three primary components: the spectroscope, the camera, the sample holder, and the light source.

The spectroscope used in this project is a diffraction grating type as shown in the Figure 33, commonly employed in gemstone identification due to its simplicity and effectiveness in separating light into its spectral components as mentioned in Theory section. However, due to commercial confidentiality, the retailer does not disclose detailed specifications or design files for this component and the component is unable to disassemble for measurement.
For image capture, a Raspberry Pi camera module equipped with a 5MP OV5647 sensor was selected, shown in Figure 34. This is paired with a 16 mm fixed focal length 1080P 1/3'' M12 mount lens. This combination was found to provide the best resolution and clarity for capturing spectral data. Several alternatives—including other M12 lenses and the Raspberry Pi High Quality (HQ) camera—were evaluated (shown in Appendix 1), but none achieved the same level of performance.

Figure 33 Conventional Grating Spectroscope


Figure 34 Raspberry Pi CSI camera


Figure 35 M12 mount 16mm focal length 1080P 1/3'' lens

Since this project focuses on visible-band absorbance spectrometry, a broadband light source is essential to ensure accurate spectral measurements across the 400–700 nm range. After evaluating several options, a parallel white light LED source was selected as shown in Figure 36.
White LEDs offer a broad emission spectrum typically ranging from 360 nm to 900 nm, making them highly suitable for visible light applications, with low initial and operating costs, minimal warm-up time, and no need for thermal regulation. Furthermore, the selected light source includes an integrated power adapter, which allows manual adjustment of light intensity—offering additional flexibility during calibration and measurement. Compared to other broadband alternatives such as tungsten-halogen or xenon light sources, which demand active temperature control to prevent overheating [33].

Figure 36 Parallel white LED light source

In conventional spectrophotometer setups, the optical path is typically linear—light travels in a straight line through the sample and directly into the spectrometer, where it is diffracted and captured for analysis. While effective, this arrangement requires a relatively long structure to accommodate the straight-line path, which can be limiting in space-constrained applications.

To address the compactness requirement of this project, a U-shaped optical path was developed. This design introduces two 90-degree redirections in the light path by incorporating two 45-degree coated full-reflection K9 glass prisms. As illustrated in Figure 37, parallel light emitted from the source is first reflected by the prisms, directed through the sample, and then goes into the spectroscope.

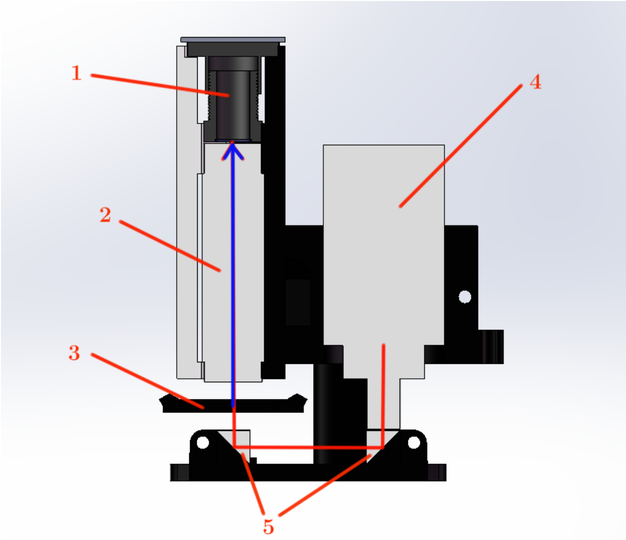
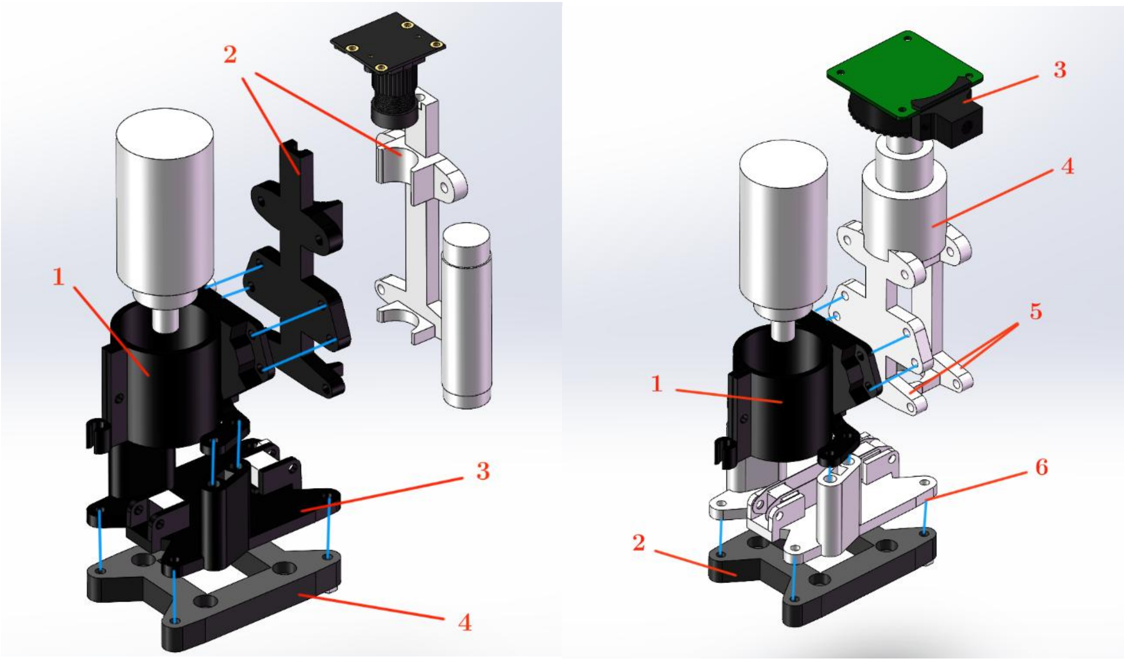


Figure 37 Illustration of optical path, 1. CSI camera, 2. Spectroscope, 3. Sample holder, 4. Parallel light source, 5. 45-degree reflection prism

As emphasized throughout the project, modularity is a core design principle intended to support flexibility, rapid iteration, and future upgrades. This modular approach is fully reflected in the spectrometer assembly, where all custom-designed structural components are created with interchangeability in mind.

Each major element of the spectrometer—including the camera, spectroscope, light source, and optical prisms—is mounted using independent, dedicated holders. These holders are designed to interface with a standardized set of assembly points, ensuring consistent alignment regardless of which specific components are installed. This means that any single module can be replaced or upgraded without requiring significant redesigns of the surrounding system.

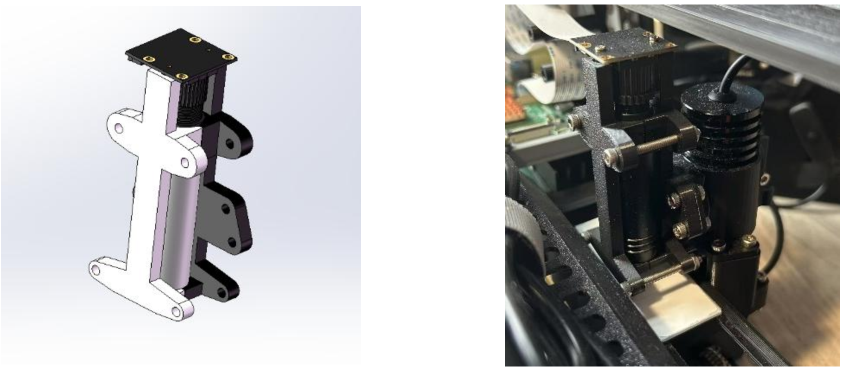


(a) (b)

Figure 38 Illustration of Modular design, a1 and b1 are the same light source holder, a2 is the clamp mechanism used for camera, spectroscope assembly, a3 is the V1 prism base, a4 and b2 are the same adapt base used for connecting the spectrometer with the linear moving platform, b3 is the Raspberry Pi HQ camera, b4 is the M12 mount 6-22m manual adaptive focal length 5MP 1/2.5'' Lens, b5 is the clamp mechanism for the HQ camera set. b6 is the V2 prism base.

For instance, as demonstrated in the Figures 38, the components labelled 1 and 4 in the left figure are identical to components 1 and 2 in the right figure. Despite differences in the surrounding parts, the common mounting interface allows seamless integration and compatibility.

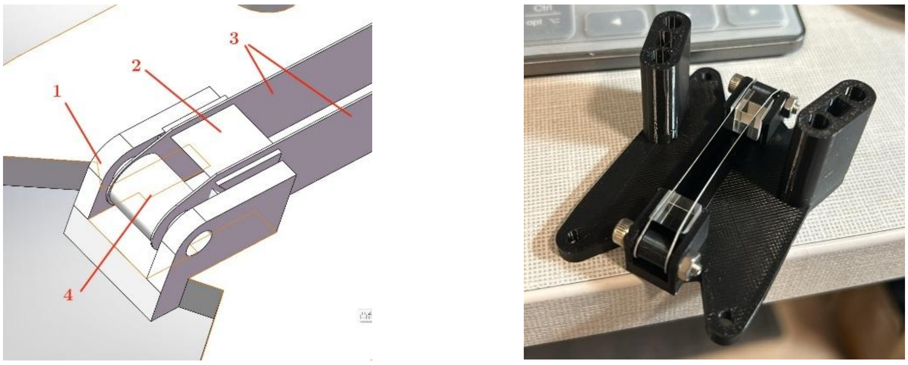
This modular architecture not only accelerates prototyping and development by enabling targeted component swaps but also ensures long-term adaptability. As requirements evolve or new optical or imaging components become available, they can be integrated into the system with minimal structural modification—making the design highly scalable and future-proof.



(a) (b)

Figure 39 Illustration of clamping structure, a. Design file, b. Real asseembly

In the current design, the camera and spectroscope are secured together using a single clamping mechanism, as illustrated in Figure 39. This design decision was primarily made to minimize optical path misalignment, which can occur due to cumulative assembly tolerances when the two components are mounted separately. By fixing both elements within a single clamping structure, their relative alignment is preserved with greater consistency, ensuring more accurate and repeatable spectral capture. Represents a deliberate trade-off between modularity and measurement accuracy, prioritizing optical performance in a critical section of the system.



(a) (b)

Figure 40 Illustration of Prism positioning mechanism, a. Design file, 1. Prism holder base, 2. Prism, 3. Laser cut metal plate, 4. 3D printed support to keep spacing of two metal plate, b. Real Assembly

In line with the alignment considerations made for the spectroscope and camera, the precise positioning of the two prisms used to bend the optical path also required additional considerations. In early design iterations, the prism holders were entirely 3D printed and relied on vertical walls with 45-degree slopes on inner side of the wall on side of prism to fix their orientation, as shown in Figure 41. While this method was straightforward and cost-effective, it proved insufficient in maintaining precise optical alignment. Due to the inherent limitations of FDM 3D printing—particularly in surface flatness and dimensional accuracy of the layered slope—the sloped surfaces failed to hold the prisms in the exact orientation required for consistent light reflection.

To address this issue, the final design is a hybrid solution. Two laser-cut metal plates were introduced to constrain both the position and angular alignment of each prism with higher precision, as shown in Figure 40. This adjustment significantly improves the stability and accuracy of the optical path, ensuring that the light is properly redirected while maintaining the system’s compact layout.

Figure 41 Early version of the prism holder.

5.2.3 Structural design

As the goal of this project is to develop a fully portable spectrophotometer system, careful planning of the power supply for each component is essential—not only from an electrical standpoint but also in terms of structural integration. Multiple subsystems require individual power sources, including the stepper motor and its driver, the adaptive light source, the Raspberry Pi 4B, and the display screen.

Importantly, power supplies and their associated cables occupy considerable physical space within the device housing. Therefore, their placement must be thoughtfully considered during the structural design phase to avoid interference with moving parts, maintain compactness, and ensure ease of maintenance. The physical footprint of each power module, along with heat dissipation and cable routing, directly impacts the overall layout and portability of the system.

The power input requirements for each component are summarized in the following chart, serving as a foundation for both electrical design and mechanical arrangement:

Chart 2 Component power specification list

Components	Input voltage	Power consumption
Adaptive light source	0-5V DC need able to adapted Direct positive and ground power cable	Max 5W
TB6600 stepper motor drive	Select from 12V-24V DC Direct positive and ground power cable	If set to 3.5A setup the consumption would be 42w. The full setup is shown in Figure
Raspberry Pi 4B	5V DC Type-C input	3A minimum. Under Typical load, around 6.4W
10.1 inch Screen	5V DC micro USB input	2A, around 10W

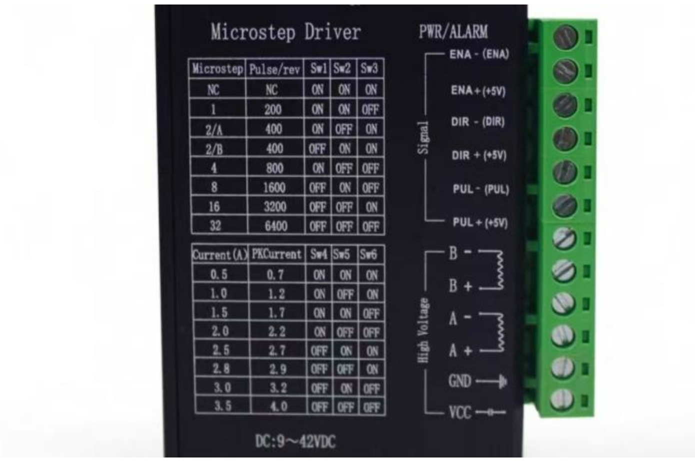


Figure 42 TB6600 setup selections

As shown in the chart, each component in the system has distinct power requirements—not only in terms of voltage and current specifications but also in connector types and interface standards. Designing a custom power control board capable of delivering multiple outputs, including 0–5V adjustable voltage, fixed 12V, 5V/3A, and 5V/2A USB outputs, would introduce significant complexity and time consumption during the development phase.

Based on the power and component requirements outlined previously, and drawing inspiration from conventional medical device layouts, the internal structure of the system has been strategically organized for both functionality and space efficiency. The power supplies and adapters are positioned at the rear of the device, isolating heat-generating elements and keeping bulky components away from user-facing interfaces. The screen display is located at the front for ease of operation, while the linear motion platform and spectrometer are centrally positioned to maintain balance and structural symmetry.

Since the stepper motor is mounted on one side of the linear platform, cable routing and port accessibility also played a key role in component placement. To allow the Raspberry Pi's ports to be easily accessed from the external shell, and to simplify connections to the stepper driver, both the Raspberry Pi 4B and the TB6600 driver are positioned on the right side of the device. This layout not only reduces internal wiring complexity but also enhances user convenience during setup and maintenance shown in Figure 43.

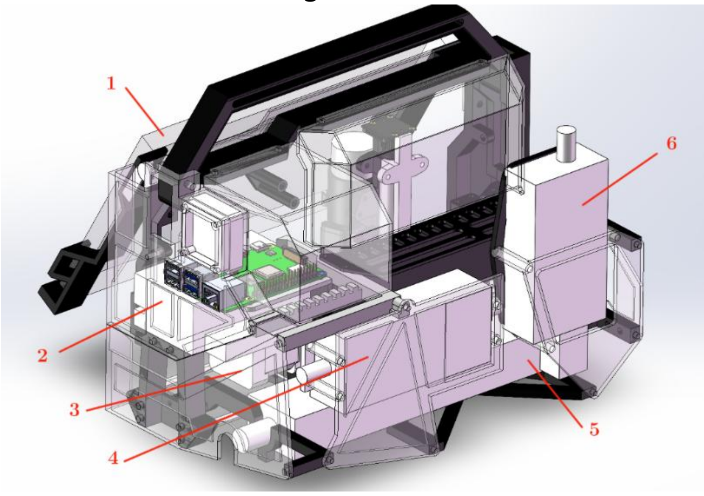


Figure 43 Illustration of component placemen, 1. 10.1 inch screen, 2. TB6600, 3. Stepper motor, 4. TB6600 power adapter, 5. Power stripe 6. Light source power adapter


Figure 44 Real assembly indication

Due to the rapid and continuous iterations undertaken throughout the development process to explore and evaluate various solutions, the structural design of the device has been intentionally made highly modular. As shown in Figure 45, the main frame incorporates numerous alternative mounting and assembly points, providing maximum flexibility for component adjustments.

For instance, all external shielding components are designed to be interchangeable, making it easy to accommodate changes in hardware dimensions or configurations as new components are introduced during prototyping or future upgrades.

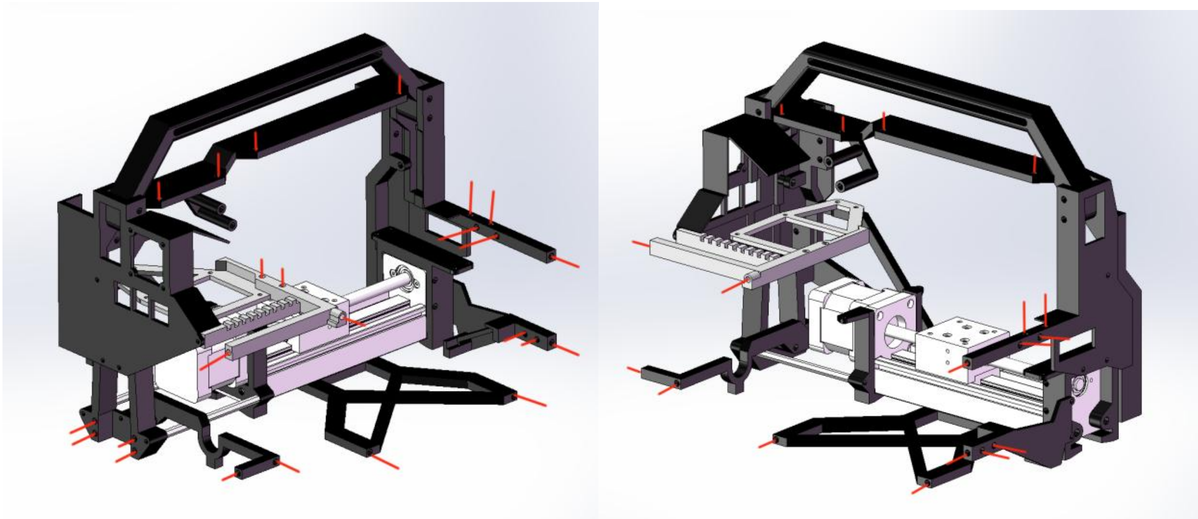


Figure 45 Illustration of alternative mounting and assembly points

To maximize structural strength and durability, all self-designed 3D printed components were carefully oriented during slicing based on their expected load-bearing directions. Particular attention was given to aligning the printing direction perpendicular to the primary stress direction, ensuring that the parts perform optimally under mechanical load.

As illustrated in Figure 46. Due to the anisotropic nature of Fused Deposition Modeling (FDM), 3D printed parts typically exhibit significantly lower strength along the Z-axis (vertical direction) compared to the X-Y plane (horizontal). Therefore, components subjected to vertical loads were printed such that these forces align with the more resilient horizontal layer orientation. This strategic approach enhances resistance to delamination and structural failure.

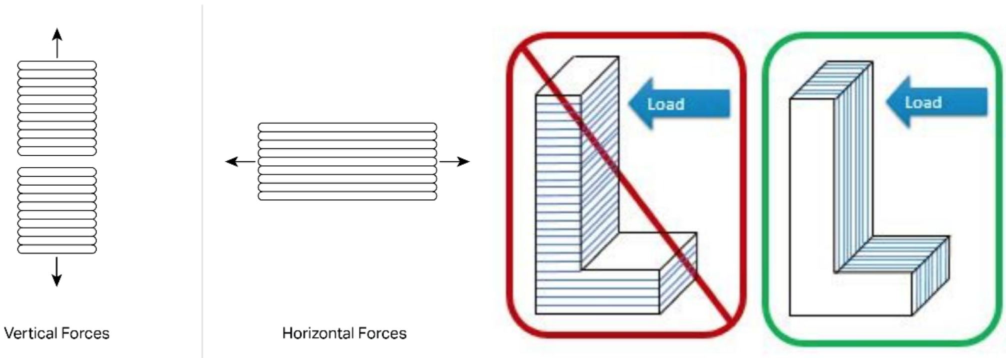


Figure 46 Illustration of how force should be loaded on 3D printed parts [34][35]

In addition to optimizing print orientation for structural strength, all 3D printed components were intentionally designed with a flat "bottom surface" to enable support-free printing, as illustrated in Figure 47. The result is a set of well-finished, functionally reliable components that require minimal post-processing and integrate seamlessly into the overall assembly.

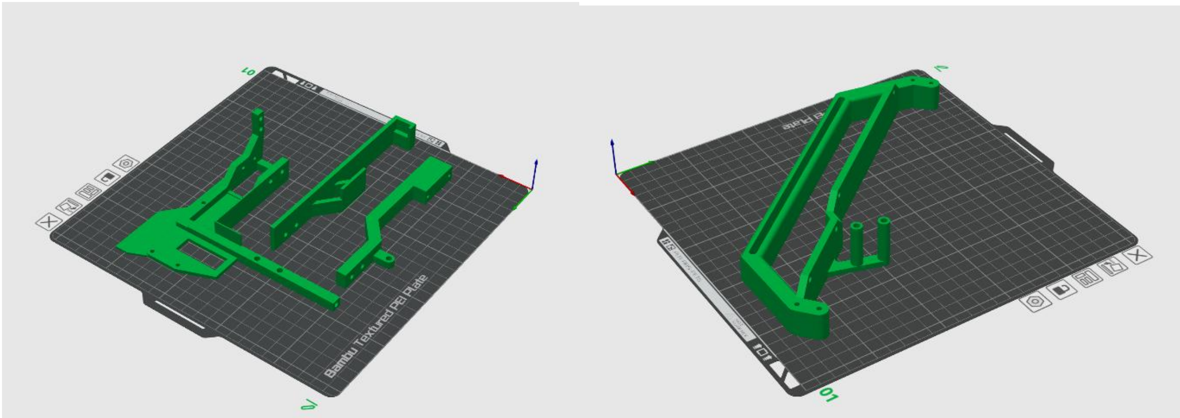


Figure 47 Example of “bottom side” consideration in 3D printer slicing application

5.3 Software Design

Similar to the Hardware Design, the Software Design is also structured into three main components, following the overall system architecture outlined in the General System Design section. These components include the control of the linear moving platform, the operation and data handling of the spectrometer, and the development of the user interface. As illustrated in Figure 48, the software system is organized into three main Python files: Main_gui.py, which serves as the central user interface, and two supporting modules—Platform_program.py for controlling the linear platform and Spectrometer_program.py for managing spectrometer setup and image capture. The application is developed using the PyQt5 framework, which provides a flexible and robust architecture for GUI-based applications. PyQt5 offers an intuitive interface layout system and a wide range of ready-to-use widgets, simplifying the creation of a professional and user-friendly UI. Additionally, PyQt's signal-slot mechanism enables safe and efficient communication between different threads and modules, allowing for reliable cross-process variable transmission. This makes it particularly suitable for integrating real-time functionalities such as camera image display, stepper motor control and live spectrum plotting within the same interface as this project is intended. In this section, each software component will be explained in detail, focusing on its specific role and implementation within the PyQt5 architecture.

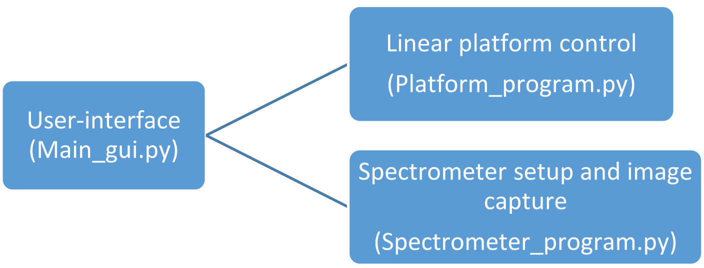


Figure 48 General file structure

5.3.1 Linear moving platform control

As described in the system design section, the Raspberry Pi controls the stepper motor through a TB6600 stepper motor driver. The connection setup is illustrated in Figure 49. In this configuration, GPIO pin 35 on the Raspberry Pi is used for the direction control signal (DIR), while pin 38 is assigned for the step pulse (STEP). The TB6600 driver receives these signals to control the motor's rotation direction and stepping sequence. The driver outputs phase currents to the stepper motor through four terminals labeled A+, A–, B+, and B–, which correspond to the two motor windings. This setup ensures reliable and precise control of the motor's position and speed during system operation.

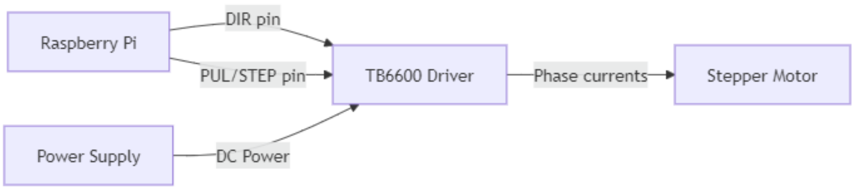


Figure 49 Connection indication for motor control

As mentioned in the previous section, most stepper motor control functions are capsulated in the Platform_program.py file as a QObject – PlatformController. Key responsibilities of the Platform Control Module include:

1. Precise stepper motor control with variable speed and direction
2. Endstop detection for safety and position reference
3. Non-blocking operation through multi-threaded design
4. Position tracking and management
5. Safe homing operations to establish reference positions

Main functions are as follows:

- **check_endstop()**

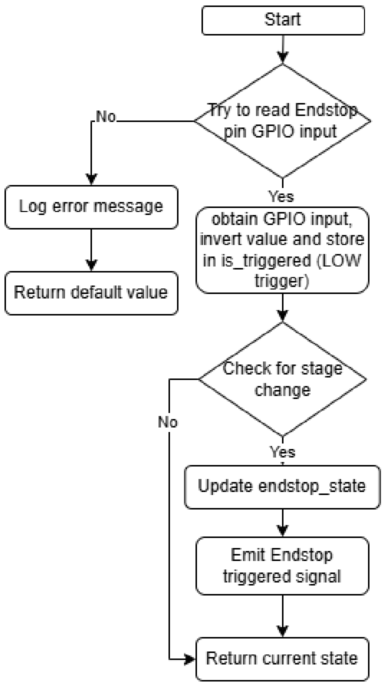


Figure 50 Flowchart for check_endstop()

The design of the endstop detection function incorporates several key considerations to ensure reliability and maintain a clean separation between hardware control and the user interface. Firstly, the function uses inverted GPIO input logic due to the endstop's normally-open configuration paired with a pull-up resistor, which causes it to read LOW when triggered. To avoid overwhelming the UI with redundant information, the function is designed to emit signals only when a state change is detected. Finally, instead of making direct function calls, the function communicates state changes to the UI via PyQt signals, ensuring a modular and decoupled system architecture.

- **Motor Movement Public-Private Function Pair, move_steps() & _move_steps_thread()**
  The motor control system employs a carefully designed public-private function pair approach that separates the movement API from the actual movement execution. This architecture is fundamental to the system's non-blocking design. The general execution flow is as shown in the Figure 51

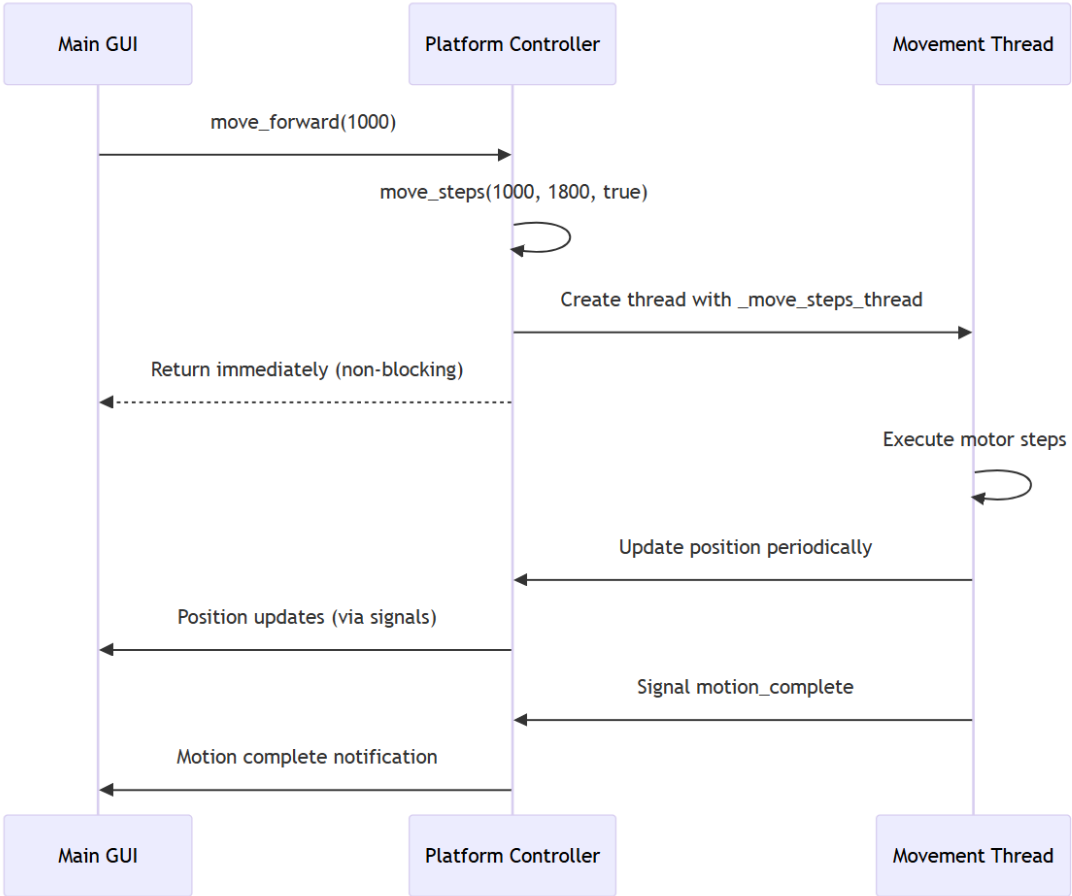


Figure 51 Execution flow for the motor control

move_steps()

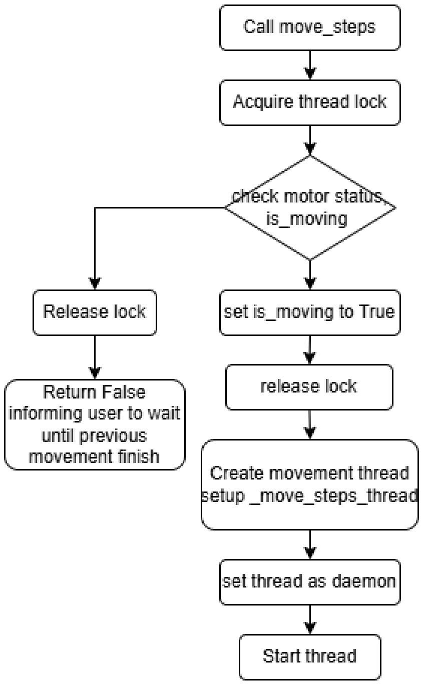


Figure 52 Flowchart of move_steps() function

The move_steps() function acts as the public interface for initiating stepper motor movement requests, typically triggered by external components such as the main GUI. As illustrated in the Figure 52 the function first acquires a thread lock and checks whether the motor is already active by inspecting the is_moving flag. If the motor is currently in motion, it releases the lock and returns False, signaling to the user that a new movement cannot be initiated until the previous one completes. Otherwise, it sets the is_moving flag to True, releases the lock, and proceeds to create a new daemon thread to handle the actual movement process through a private function. By offloading execution to a separate thread, move_steps() ensures a non-blocking response, maintaining responsiveness in the main application while effectively managing motor state and execution concurrency.

_move_steps_thread()

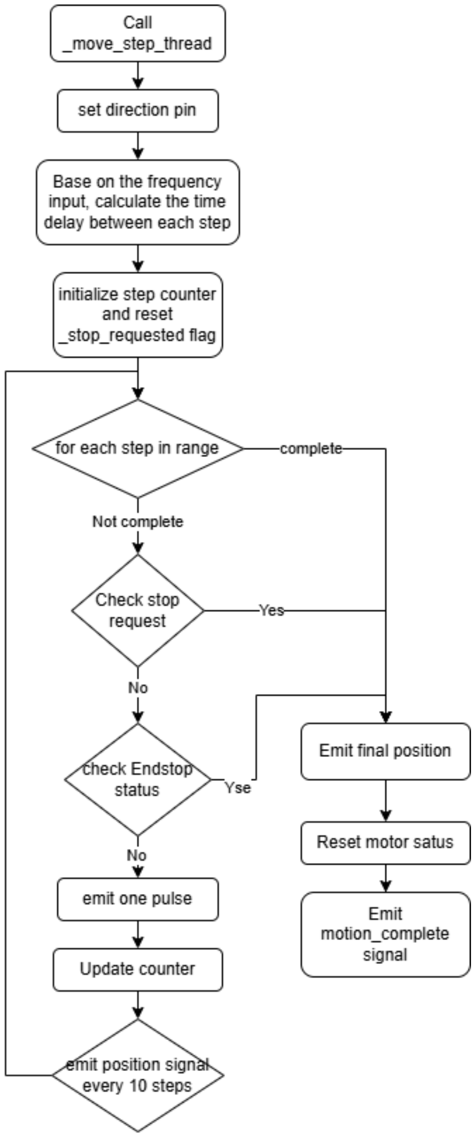


Figure 53 Flow chart of _move_steps_thread()

The _move_steps_thread() function encapsulates the core motor control routine and is designed to run in a separate thread, allowing it to handle time-sensitive operations independently of the main application thread. As depicted in the flowchart, the function begins by setting the motor direction through GPIO control and calculating the delay between pulses based on the specified frequency. It then initializes a step counter and clears any prior stop requests. The function enters a loop to execute each movement step, checking on each iteration whether a stop has been requested or if the endstop has been triggered. If neither condition is met, it issues a single pulse to the TB6600 driver and updates the position counter accordingly. For improved responsiveness and UI feedback, a position signal is emitted every 10 steps. Upon completing the movement or encountering a stop condition, the function emits the final position, resets the motor's status flags, and sends a motion_complete signal to notify the UI. Through careful timing management, real-time monitoring, and safe GPIO interaction, _move_steps_thread() ensures reliable, accurate, and non-blocking stepper motor operation.

For each pulse, which commands the motor to take a single step, consists of a high-low signal cycle applied to the STEP pin connected to the TB6600 driver. The timing of this cycle is determined by the desired stepping frequency, using the following formula:

$$Delay = \frac{1}{2 \times f}$$

Where f is the target frequency. For each pulse has a rising edge and a falling edge, thus a full pulse need two time delays.

- **Homing Calibration Public-Private Function Pair:**

  home()

  The home() function acts as the public interface for initiating the motor's calibration or zeroing procedure, following the same non-blocking, thread-based architecture used in move_steps(). The function begins by enforcing a fixed homing direction—typically negative—ensuring that the motor moves toward the endstop for consistent and safe zeroing. It then acquires a thread lock to check whether the motor is already in motion by evaluating the is_moving flag. If another operation is active, the function releases the lock and informs the user that homing cannot proceed until the current action is completed or interrupted. Otherwise, it updates the motor's state by setting the is_homing flag, releases the lock, and spawns a dedicated thread to handle the actual homing logic. This design keeps the main application responsive while isolating time-sensitive or hardware-level operations within a background thread, preserving modularity and robustness in the control system.

  _home_thread()

  The _home_thread() function carries out the actual homing routine, handling the detailed and time-sensitive execution steps required to calibrate the motor's zero position. It repeatedly moves the motor in the negative direction while checking whether the endstop is triggered, the maximum step limit is reached, or the operation is manually cancelled. Once the endstop is successfully engaged, the function verifies if the motor has indeed stopped, logs a successful homing event, and then performs a back-off motion in the positive direction to relieve pressure on the switch. The motor’s position is then reset to zero, and the system marks the homing operation as successful. In the event of a failure (e.g., if the endstop isn't triggered properly), it logs the error accordingly. Regardless of the outcome, the function concludes in a finally block by clearing the is_homing flag and emitting a homing_complete signal to notify the UI. This threaded design allows the homing process to run safely and asynchronously, maintaining system responsiveness while ensuring precise and reliable calibration.

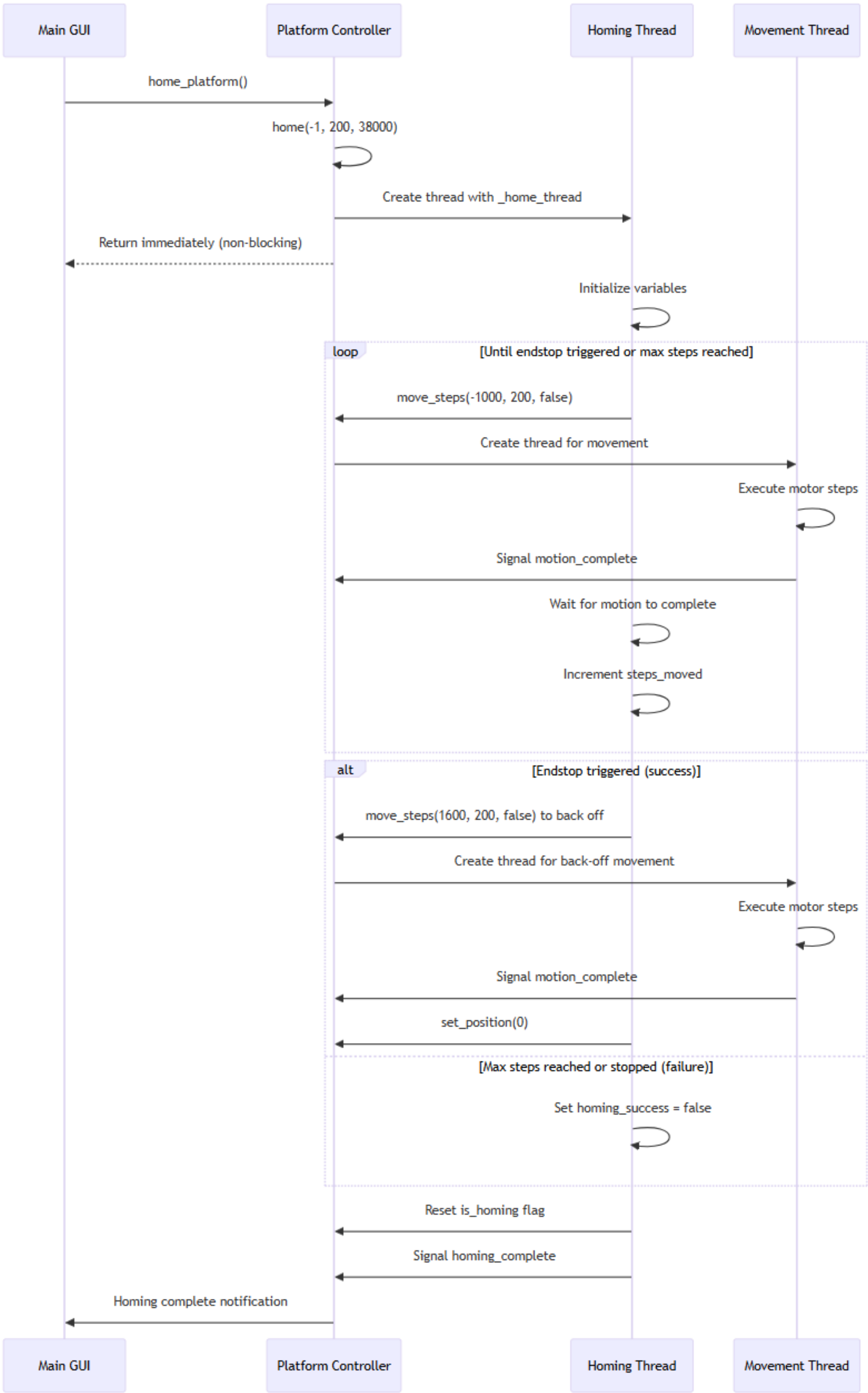


Figure 54 Execution flow for the homing process

5.3.2 Spectrometer display

The Spectrometer_program.py module serves as the core component for spectrum acquisition and analysis in the spectroscopy system. This module provides a comprehensive interface between the Raspberry Pi camera system and the main application, enabling real-time spectrum capture, analysis, and visualization.

Key responsibilities of the Spectrometer Control Module include:

1. Camera initialization and management with optimized resolution settings
2. Continuous frame capturing in a non-blocking thread-based architecture
3. Real-time spectrum extraction from camera frames
4. Wavelength calibration and mapping
5. Spectrum analysis including peak detection and filtering
6. Color representation of spectral data
7. Signal transmission to update the user interface in real-time

Main function are as follows:

- **_setup_camera()**

  The function logic is straightforward, but the setup for view frame and the camera input is worth mentioning, while having great impact on general performance and spectrum result. The _setup_camera() function initializes the imaging system by connecting to the Raspberry Pi camera, configuring the colour format to RGB888, and establishing a separate thread for continuous frame capture similar to the stepper motor control architecture.

  Early development attempts utilized the camera's maximum resolution (up to 3280×2464 pixels), which provided extremely detailed spectral images but resulted in severe system lag. The current implementation deliberately selects a moderate 1600×1200 resolution that delivers sufficient spectral detail while maintaining smooth system operation.

  RGB888 was chosen for the camera configuration to balance color depth and processing efficiency for spectral analysis. Unlike YUV or JPEG, it avoids extra conversion steps and potential artifacts. RGB888 allows direct access to red, green, and blue channels, aligning with the system's weighted spectral extraction method and simplifying the downstream quadratic polynomial fitting process.

  This resolution optimization significantly enhances both system performance and spectral quality. The controlled frame rate (30 FPS) ensures consistent timing between spectral samples, critical for accurate intensity measurements. The thread-based design prevents camera operations from blocking the user interface, allowing simultaneous interaction and data collection.

- **_read_calibration()**

  Because the grating could cause the light spectrums to be diffracted unequally as mentioned in the Theory section, the program would need to remap the spectrum to the pixels in order to obtain correct spectrum intensity, taking inspiration from the opensource project PySpectrometer2 project, the calibration method is as follows

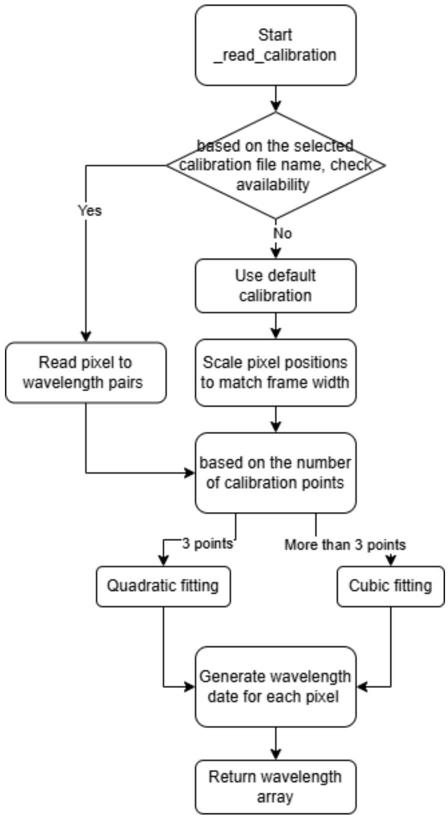


Figure 55 Calibration method flowchart

Based on the number of calibration spectrums can be provided, the function is able to perform quadratic fitting and cubic fitting to obtain the spectrum to pixel mapping. Take 3 spectrums as example, based on this three given pairs, it is able to be used to solve a quadratic polynomial:

$$\lambda(p) = a \cdot p^2 + b \cdot p + c$$

Where $\lambda(p)$ is the wavelength at pixel position p

a, b, and c are coefficients determined by the calibration data

By using np.polyfit(pixels, wavelengths, 2) function, which calculates these coefficients by solving a least-squares problem, minimizing the sum of squared differences between the known calibration points and the polynomial model predictions.

Once the coefficients are determined, the np.polyval(coeffs, pixel) function applies the polynomial function to each pixel position from 0 to the image width, generating a complete mapping array. This ensures that every pixel column in the captured image can be associated with a specific wavelength value. The resulting wavelength data array serves as a lookup table for all subsequent spectral analysis, enabling the conversion from spatial information in the camera image to spectral information in nanometers.

- **capture_spectrum()**

  The capture_spectrum() function is used to process the camera footage and produce the intensity array, implementing a dynamic weighting mechanism that adaptively adjusts the contribution of each color channel (RGB) based on wavelength regions. This approach is intended to enhances the accuracy of spectral extraction across the visible spectrum. The general process is as shown in Figure 56

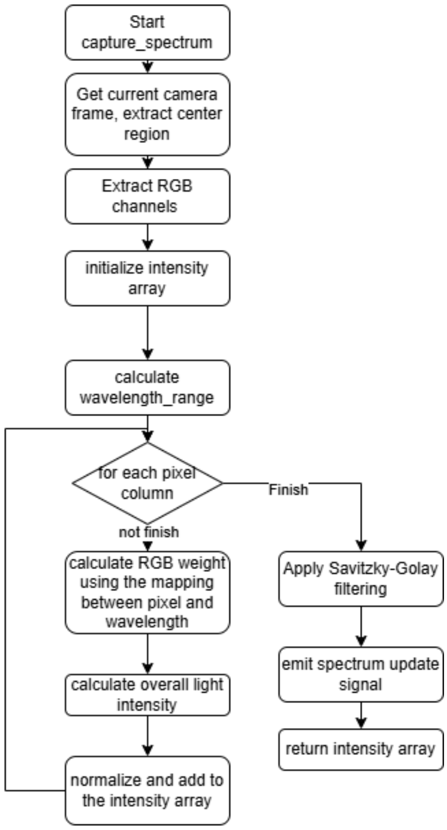


Figure 56 Flowchart for capture_spectrum()

The weighting system employs a normalized wavelength approach where each pixel's wavelength is first converted to a normalized value between 0 and 1:

norm_wl = (wavelength - wavelength_min) / wavelength_range

Based on this normalized value, three specialized weighting functions are applied to determine how much each RGB channel should contribute to the final intensity value:

kr = 0.1 + 0.8 * norm_wl2**

This is a quadratic function that increases dramatically as wavelength increases, giving the red channel maximum influence in the long wavelength (red) region of the spectrum.

kg = 0.1 + 0.8 * 4 * norm_wl * (1 - norm_wl)

This is a quadratic function that peaks in the middle of the wavelength range (green region), following a parabolic distribution peaking at norm_wl = 0.5.

kb = 0.1 + 0.8 * (1 - norm_wl)2**

This is a quadratic function that decreases as wavelength increases, giving the blue channel maximum influence in the short wavelength (blue) region of the spectrum.

Each weighting function maintains a minimum weight of 0.1 (10%) regardless of wavelength, ensuring that every channel always contributes at least partially to the spectral data. The remaining 0.8 (80%) of the weight is distributed according to the quadratic functions.

v = r_sum * kr + g_sum * kg + b_sum * kb

Finally the light intensity v at this pixel can be calculated by this formula, which adding up RGB channel intensity by their corresponding factors. This procedure is mainly used to addresses a significant limitation in conventional grayscale conversion algorithms. Standard RGB to grayscale conversion methods, such as those used in OpenCV's cvtColor(image, cv2.COLOR_BGR2GRAY), typically employ a fixed weighting formula:

Gray = 0.299*R + 0.587*G + 0.114*B

As mentioned in theory section, This formula, optimized for human visual perception rather than spectroscopic accuracy. When applied to spectral imaging, this conventional approach creates a substantial information loss in the blue region of the spectrum (approximately 380-490nm).

5.3.3 Scanning process

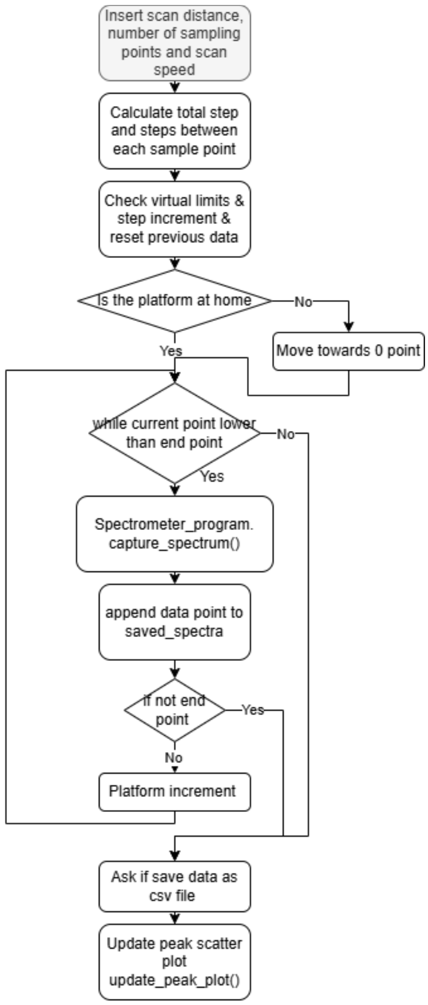


Figure 57 Flowchart for scanning process start from user interface

5.3.4 General User interface design

In order to achieve a user-friendly interface, It features a modular layout that organizes functional components logically, enabling users to easily control the platform, acquire spectral data, and perform analysis, as shown in the following Figure 58 59and 60

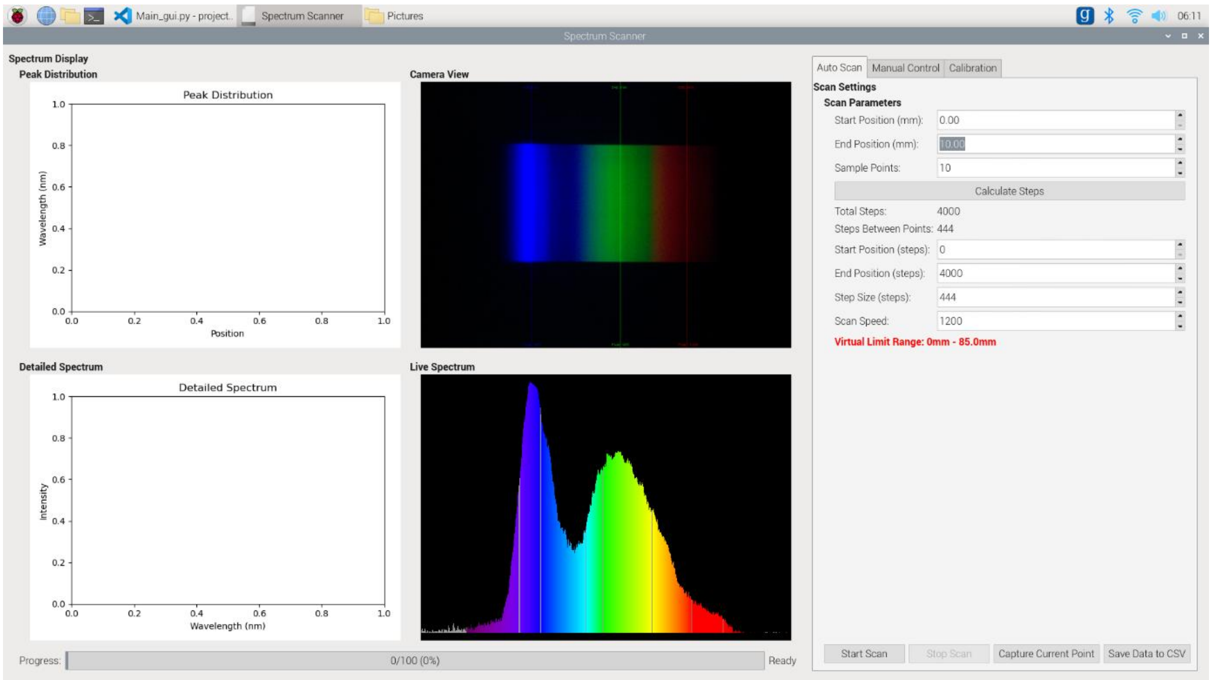


Figure 58 General view of the GUI (displaying Auto scan tab in the control area)

The main interface adopts a horizontally split layout, with the left side occupying two-thirds of the width for data display and the right side occupying one-third for the control panel. This proportional distribution ensures sufficient space for visual data while accommodating all necessary interactive elements in the control panel.

Data Display Area (Left Side)

The data display area is organized into four quadrants, containing four main components:

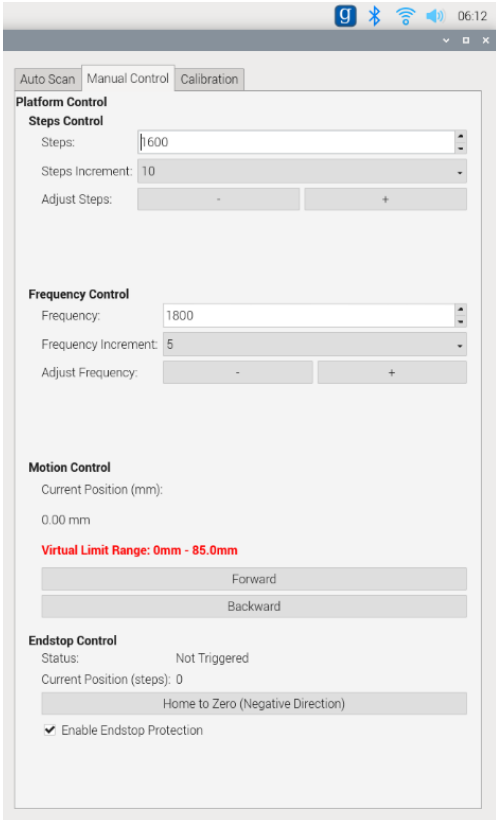
Peak Distribution Map (Top Left):
Displays all collected spectral peaks in a scatter plot format, with the horizontal axis representing position (mm), the vertical axis representing wavelength (nm), and the color of the dots indicating spectral intensity. The plot supports direct interaction—users can click on specific positions to analyze the corresponding spectrum in detail.

Camera View (Top Right):
Shows the real-time image captured by the spectrometer’s camera, with calibration line positions overlaid. In calibration mode, users can click directly on specific spectral lines in the image to mark calibration points.

Detailed Spectrum Plot (Bottom Left):
Presents detailed spectral information at a selected position, including wavelength distribution and intensity. When two points are selected, the system performs a comparative analysis and labels the primary peak wavelengths. A gradient background based on corresponding wavelengths enhances visual clarity.

Real-Time Spectrum Plot (Bottom Right):
Displays the currently acquired spectral data in color, with each wavelength rendered in its corresponding color for intuitive visualization of intensity distribution.

A progress bar and system status information are shown at the bottom to indicate the current operation progress and system state.



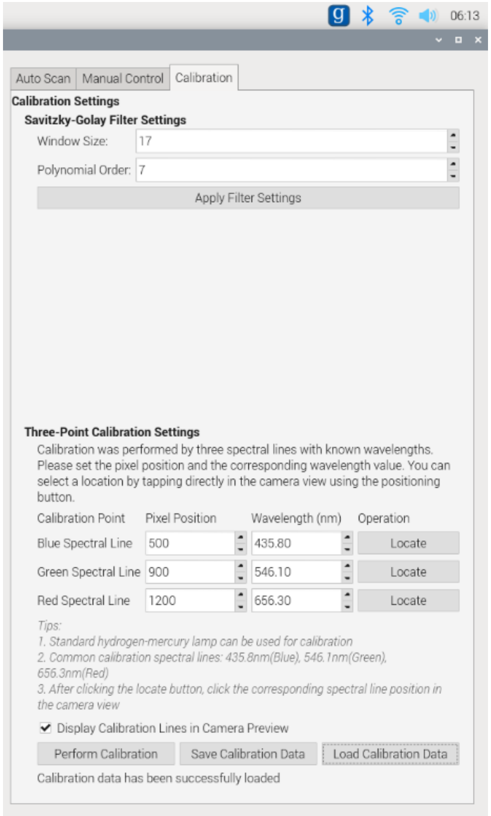


Figure 59 Manual Control and Calibration tab in control area

Control Panel (Right Side)

The control panel uses a tabbed layout and includes three main functional sections:

Automatic Scanning:

Allows users to configure scanning parameters such as start position, end position, and number of sampling points. It includes an automatic step calculation feature that computes the required steps based on physical distance. Scan control buttons and data saving options are provided at the bottom.

Manual Control:

Divided into four areas: step control, frequency control, motion control, and limit control. Users can precisely control the platform's position. A virtual limit protection system is included to prevent mechanical overrun and ensure device safety.

Calibration Settings:

Offers configuration of Savitzky-Golay filter parameters and a three-point calibration function. The calibration section displays pixel positions of characteristic spectral lines and their corresponding wavelengths. Users can click directly on the camera image to locate spectral lines, significantly simplifying the calibration process.

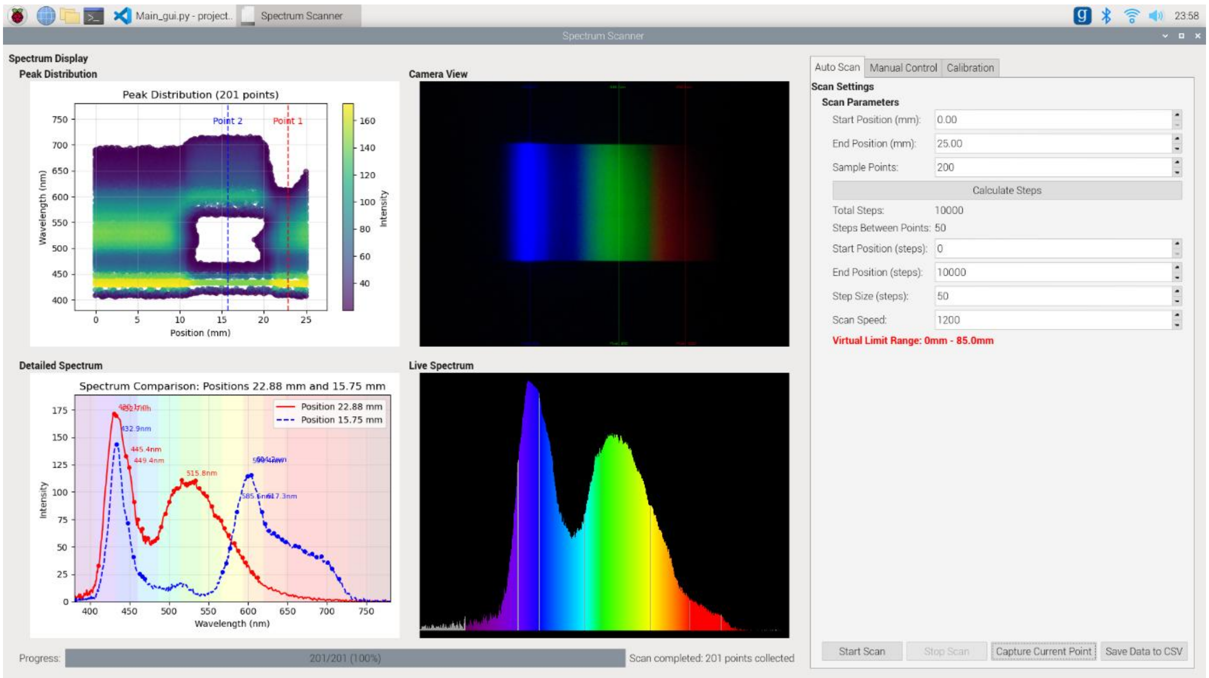


Figure 60 Example of scan result

As shown in Figure 60, the top-left section displays the peak distribution map from the linear scan, illustrating the spectral peak distribution at each sampling point. In this chart, the Y-axis represents wavelength (nm), the X-axis represents position (mm), and the colour of each dot indicates peak intensity—brighter colours correspond to higher intensities. The colour bar on the right provides an intuitive reference for the intensity mapping.

This distribution map is designed to be interactive: users can directly click on data points within the graph to select them, with up to two points selectable at the same time for comparison. The selected points are marked with vertical dashed lines, as shown in the figure with “point 1” and “point 2.” Once data points are selected, the detailed spectrum plot in the lower-left section immediately displays the full spectral curves for the selected positions—the red solid line represents the first selected point (position 22.88 mm), and the blue dashed line represents the second selected point (position 15.75 mm).

This comparison feature allows users to intuitively analyze spectral differences between different positions. The background of the detailed spectrum plot also features a gradient based on corresponding wavelengths, enhancing the correlation between wavelength and visible light colours. Key peak wavelengths are labelled with precise numerical values for accurate analysis.

6. Experimental method

To evaluate the performance and reliability of the device, comprehensive testing was conducted on its core components, including the spectrometer and the linear motion platform. The detailed experimental procedures, along with the setup and methodology for each test, are outlined in the following sections.

6.1 Linear platform test

Since high positioning accuracy was essential for the linear moving platform, a compact repeatability measurement instrument, as shown in Figure 61, was employed to evaluate its repositioning precision. The platform was first moved to a specified position, where the measurement instrument was attached to the device with slight compression. The initial reading was recorded, after which the platform was returned to its starting point. The scanning process was then repeatedly carried out to the same target position, and the readings from the measurement instrument were recorded each time to assess any changes, thereby evaluating the platform’s repeatability. Additionally, the measurement would be performed in different speed setting of the platform to check if there are step slip.

Figure 61 Repeat positioning accuracy

Additionally, the motor temperature was monitored to assess the system's thermal stability and ensure it could operate continuously over extended periods without risk of overheating. During prolonged operation, temperature readings were taken at regular intervals using an temperature sensor to evaluate whether the motor remained within safe operating limits. This test helps verify the system's long-term reliability and suitability for sustained use in real-world applications.

6.2 Spectrometer output test

To evaluate the reliability of the spectrometer, the testing procedure was divided into two main phases. The first phase involved validating the accuracy of the application by analyzing known-spectrum light sources, such as calibrated LEDs or Neon bulb with well-documented emission spectra. This allowed for a direct comparison between the spectrometer’s readings and the expected spectral profiles, providing insight into its calibration and measurement accuracy.

The second phase focused on testing a variety of real-world samples, including solid materials and the results of thin-layer chromatography (TLC) experiments. These experiments used food dyes and colored pen inks as analytes. Originally, a 7:3 hexane/acetone mixture was intended as the mobile phase; however, due to safety and handling considerations, the solvent was substituted with a 37.5% ethanol solution. This safer alternative still provided effective separation of color components on the TLC plate. The spectrometer was then used to capture the absorbance spectra across the test sample, assessing its ability to resolve and differentiate various pigments.

After collecting the spectral data, an additional experiment was conducted to reconstruct the linear color distribution of the samples from the measured spectra. This step aimed to demonstrate the system’s capability to accurately capture and digitally reproduce spatially varying spectral information, which is essential for practical applications in chemical analysis, material identification, and quality control.

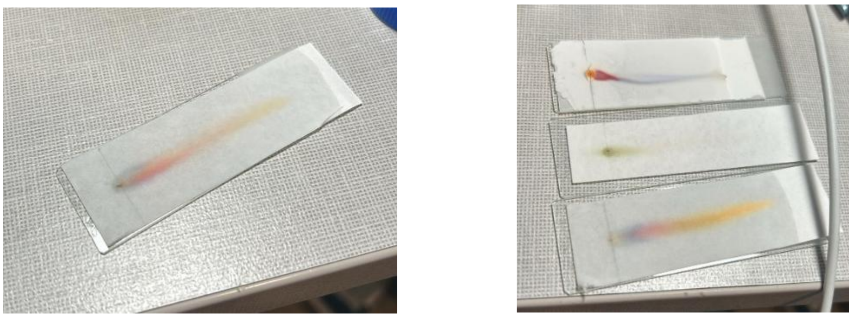


Figure 62 example of Thin layer chromatography

7. Result and Calculations

Based upon the experimental methods outlined in the previous section, this part presents the key results derived from the conducted experiments. In addition to summarizing the observed results, particular attention is given to the underlying calculations, especially the spectrometer dynamic weighting process, detailing how this approach was applied to refine the spectral data and enhance the accuracy of the analysis.

7.1 Linear moving platform performance

As described in the Experimental Method section, the accuracy of the linear moving platform was evaluated under two conditions: with and without GUI interference. The corresponding results are illustrated in Figure 63. The "without GUI" condition refers to running only the stepper motor control code using identical distance and speed parameters, without the additional overhead introduced by the graphical user interface. This comparison allows for a clearer assessment of how GUI processes may impact the precision and stability of platform motion.

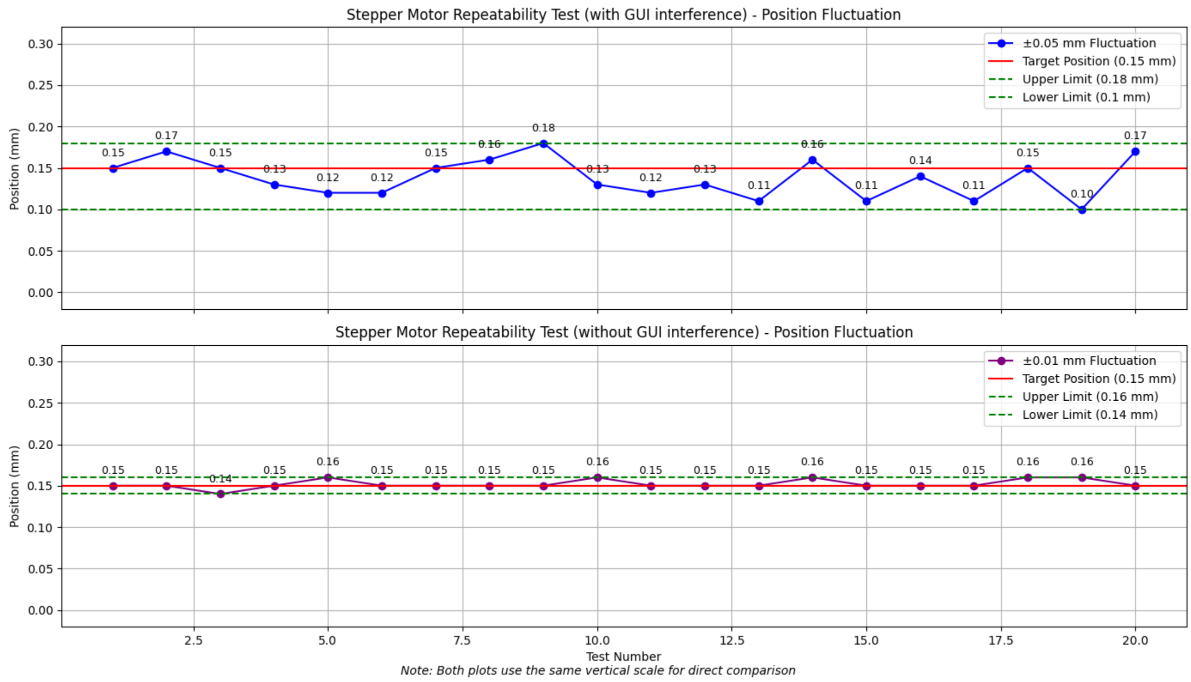


Figure 63 Comparison of linear moving platform repeatability performance with and without GUI interference

Based on the figure, it is evident that the performance of the linear moving platform is significantly affected by GUI interference, despite attempts to mitigate it using multithreading. The repeatability measurement instrument was initially calibrated to compress to 0.15 mm for both test conditions. Using the chart data, the mean position can be calculated using the following formula:

$$Mean = \mu = \frac{1}{20}\sum_{i=1}^{n} x_i$$

The result is 0.137mm for first condition and 0.152mm for second condition, while the standard deviation is calculated as follows:

$$Standard\ deviation = \sigma = \sqrt{\frac{1}{20}\sum_{i=1}^{n}(x_i - \mu)^2}$$

Based on the data, the standard deviations are 0.023 mm and 0.005 mm for the two respective conditions. Accordingly, the repeatability—defined as six times the standard deviation (6σ)—is 0.138 mm for the GUI-interfered scenario and 0.030 mm for the test without GUI interference. These results clearly demonstrate that GUI activity introduces noticeable variation in the platform's positional accuracy.
Despite the measurable impact on positional repeatability, the GUI remains functional and responsive. It is capable of controlling and interrupting the scanning process as well as initiating manual movements, with only minor delays observed during operation.

7.2 Spectrometer performance

As described in the design section, the spectral processing program incorporates a dynamic weighting algorithm to enhance the accuracy of spectrum analysis. Calibration was conducted using an RGB LED with known spectral peaks—625 nm for red, 520 nm for green, and 464 nm for blue—as shown in Figure 64. This calibration enables the algorithm to adjust for intensity variations and sensor-specific biases. The resulting visualization of the dynamic weighting effect is presented in Figure 65, demonstrating improved spectral clarity and consistency across measurements.

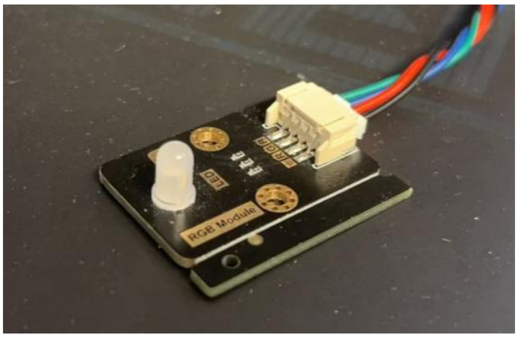


Figure 64 RGB LED

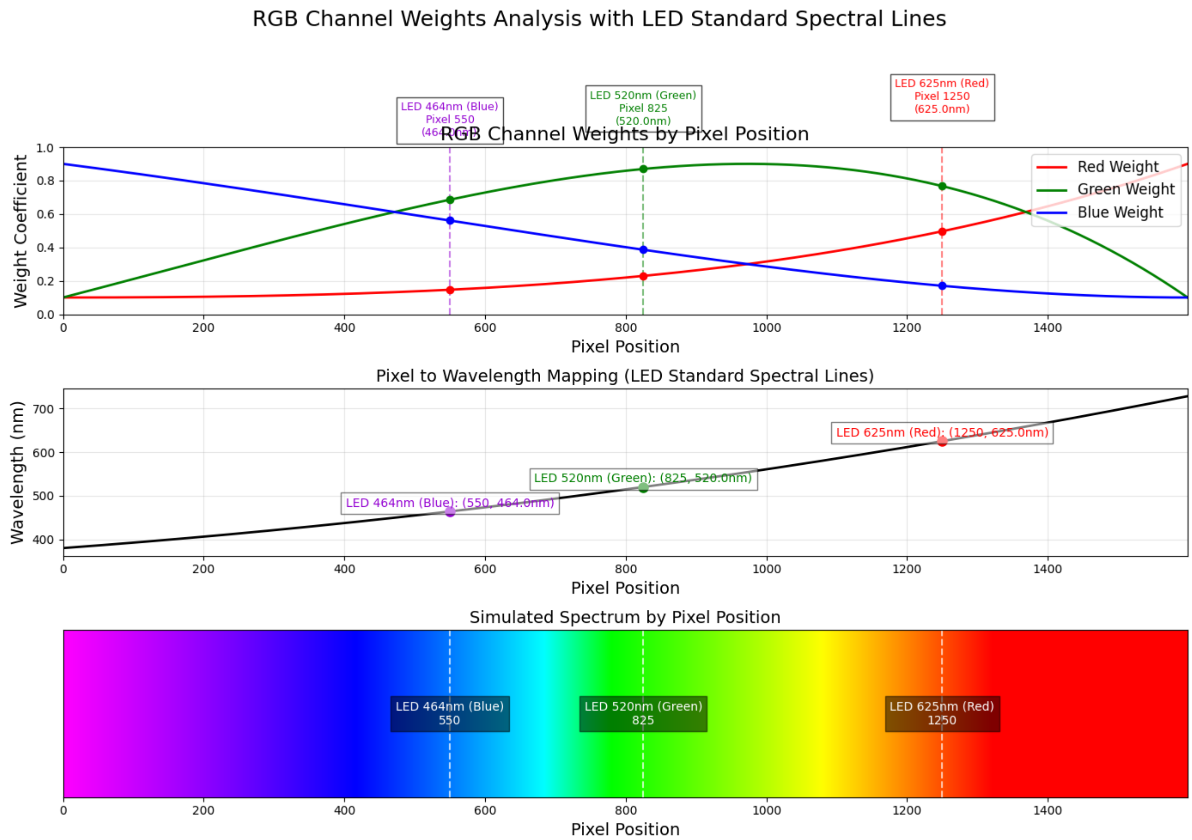
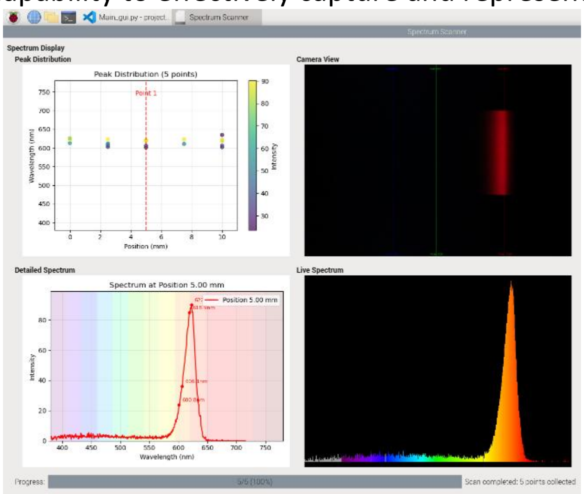
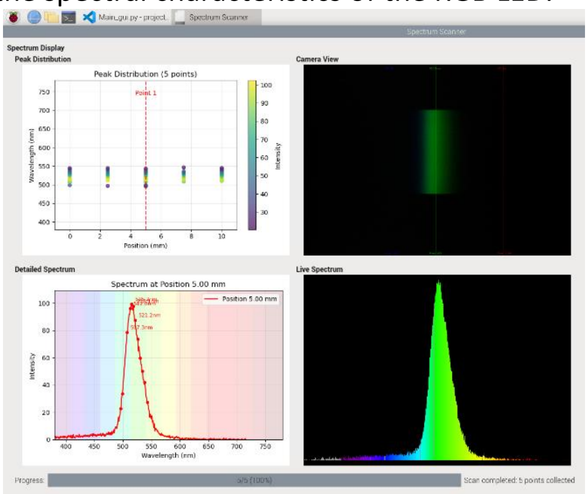


Figure 65 Resulting dynamic weighting based on the three-point calibration

The resulting spectra for the individual RGB LED components, as well as the combined spectrum, are presented below and can also be found in the appendix. As illustrated in the figures, the program successfully maps the RGB light spectrum, accurately identifying and highlighting the spectral peaks for each color component. This demonstrates the program's capability to effectively capture and represent the spectral characteristics of the RGB LED.





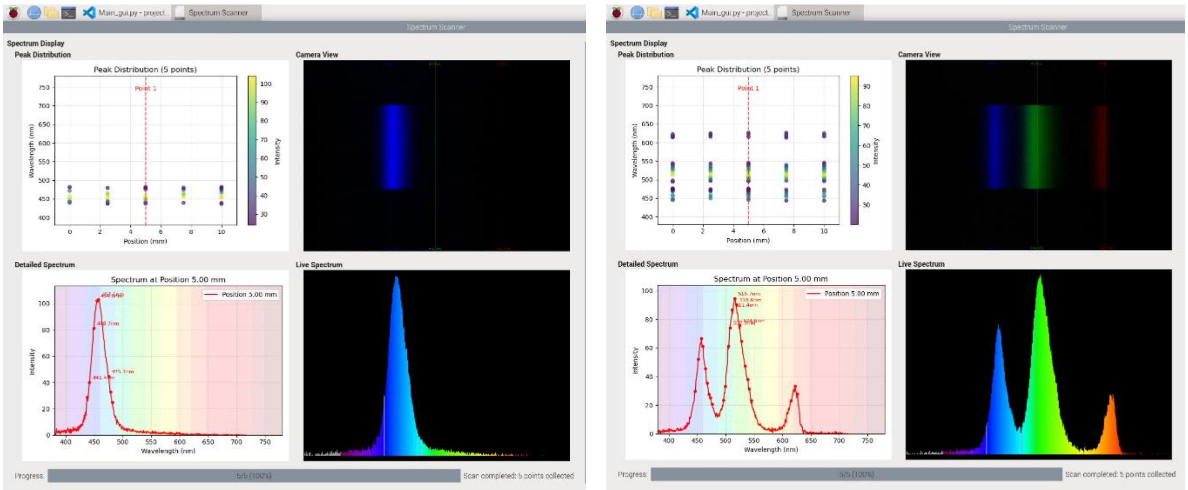


Figure 66 Spectrum results of RGB LED (also in appendix 1)

A UV LED with a specified wavelength range of 390–405 nm was also tested to evaluate the spectrometer's accuracy. The resulting spectrum, as shown below, reveals a detected peak at 400.2nm—well within the manufacturer-provided range. This result confirms the spectrometer's high precision and reliability in detecting wavelengths within the ultraviolet spectrum.

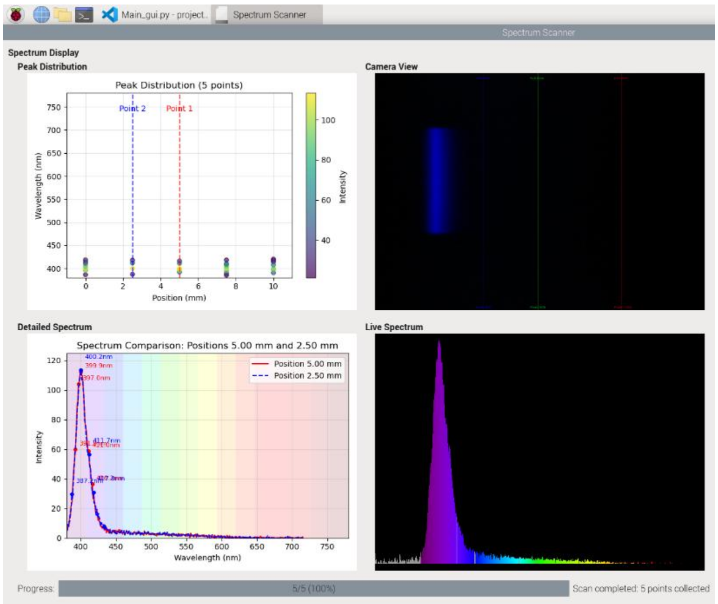


Figure 67 Spectrum result for UV LED (Also in Appendix 2)

To assess the sensitivity of the spectrometer, three 3D-printed plates in subtly different shades of yellow to refer to solid samples, were scanned using a parallel LED light source. As shown in Figure 69, both the peak scatter plot and the detailed spectral plots exhibit distinct differences among the three samples. This clearly demonstrates the spectrometer's ability to detect fine variations in colour, highlighting its high sensitivity and resolution.

Figure 68 3D printed yellow plates

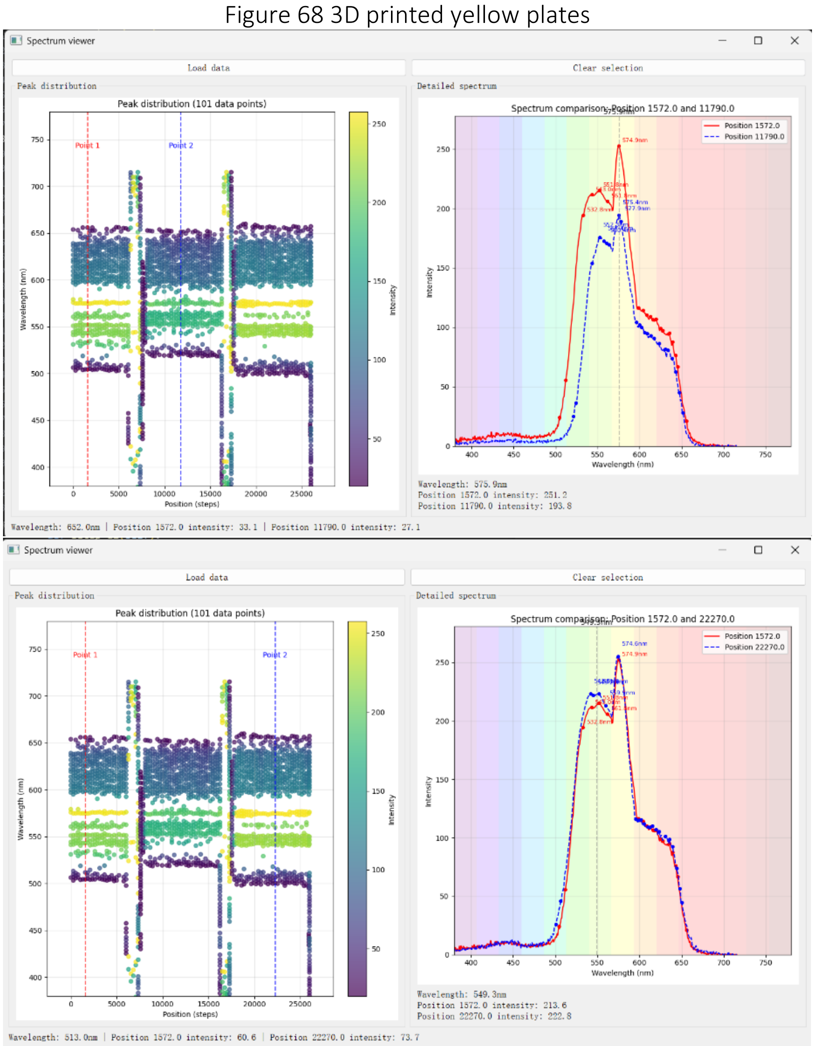


Figure 69 A reproduce from csv file of Spectrum result of the parallel light source passing through the three 3D printed plates, where the sequence is the same as shown in Figure 67. (Also listed in Appendix 3)

In addition to testing solid samples, thin-layer chromatography (TLC) result slides were also analyzed using the spectrometer. Two representative examples are shown below: the first involves the separation of compounds using an orange-coloured pen, while the second demonstrates separation using a combination of blue and red inks. These tests further validate the spectrometer's capability to analyze complex colour separations in chemical applications.

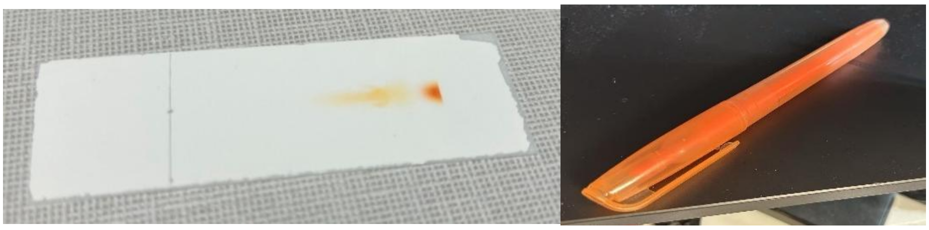


Figure 70 TLC result of orange colour pen ink

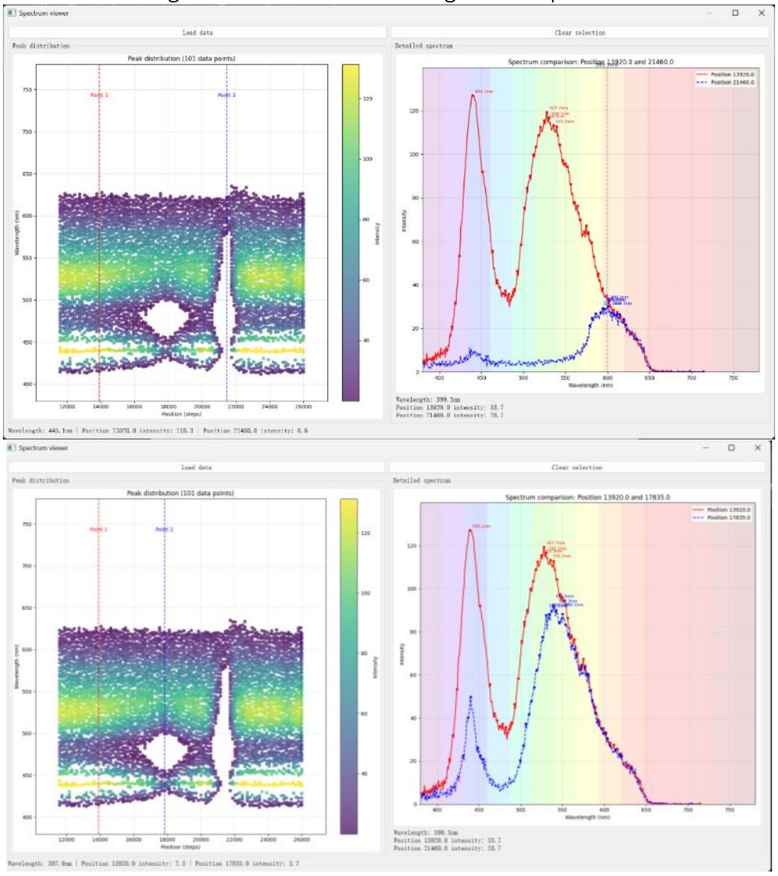


Figure 71 A reproduce from csv file of Spectrum results of scanning the Orange pen TLC slide(Also included in Appendix 4) , the two figure selected two sets of sample points for comparison, where first indicate the normal white area and the blur orange area in Figure 69, the second shows the comparison of white area with the dark orange area on the slide

Figure 72 TLC result of red and blue colour pen

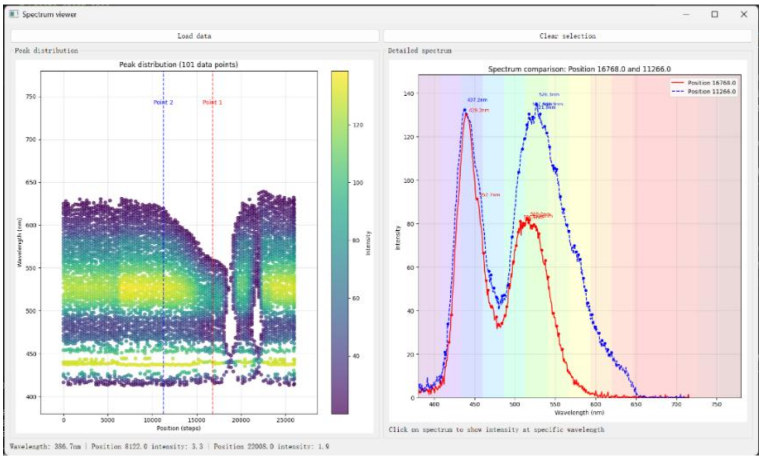


Figure 73 A reproduce from csv of Spectrum results of scanning the Red and Blue mix TLC slide (Also included in Appendix 5) the figure shows the comparison of the spectrum difference caused by the change in density of blue area shown in the Figure 71

As demonstrated by these two sets of TLC results, the spectrometer effectively captures the key characteristics of each sample, accurately distinguishing both color composition and relative density. This highlights its capability to resolve detailed variations in chemical separations.

8. Discussion

As demonstrated in the previous section, the current prototype successfully performs basic scanning functions, displays results in a user-friendly interface, and stores data effectively. The total production cost is approximately £200, and the device remains highly portable—lightweight enough to be carried single-handedly using the integrated handle. These features generally align with the original design goals of creating a low-cost, portable, modular, and user-friendly spectrophotometer. However, there remains significant room for improvement. The following section will explain on potential enhancements from three key perspectives: the spectrometer, the linear scanning platform, and the overall system design.

8.1 Possible improvements on spectrometer

As discussed in both the Design and Results sections, the system incorporates a dynamic weighting algorithm to compensate for the blue-spectrum loss typically encountered in conventional methods. However, as illustrated in Figure 74, certain limitations remain. When the light source is bright—but not intense enough to cause the spectrum to appear as a solid white line—the individual channel of the camera's CMOS sensor tends to saturate first. This saturation results in a flattened plateau in the spectral output across the corresponding region as shown in the figure.
Despite this, Figure 74 reveals a noticeable sloped curve in the area that should appear flat. This discrepancy arises because the current dynamic weighting method models the intensity changes in each color channel using a quadratic function, which does not accurately capture the non-linear behavior of the channel weight changes of different spectrums.

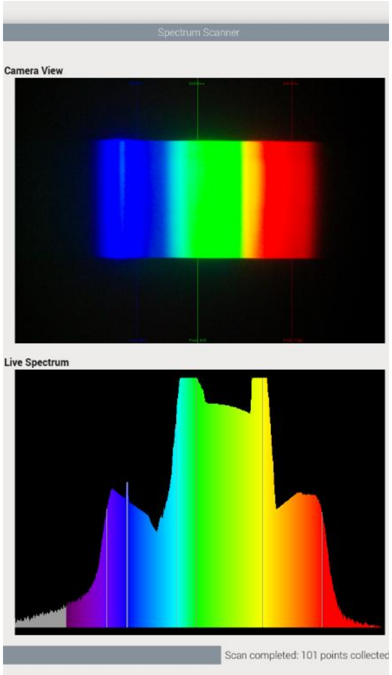


Figure 74 Saturated spectrum and stray light indication

A potential solution to this issue is to incorporate a broader set of reference spectra with known intensity distributions during the spectrophotometer's calibration process. This would allow for more accurate spectral mapping and improve the precision of the dynamic weighting algorithm. Furthermore, employing higher-order polynomials to model the channel responses could better capture the non-linear variations observed during measurements.

In addition, the accuracy of the camera-based spectral display is influenced by the inherent sensitivity of the camera's CMOS sensor, as demonstrated in Figure 75.

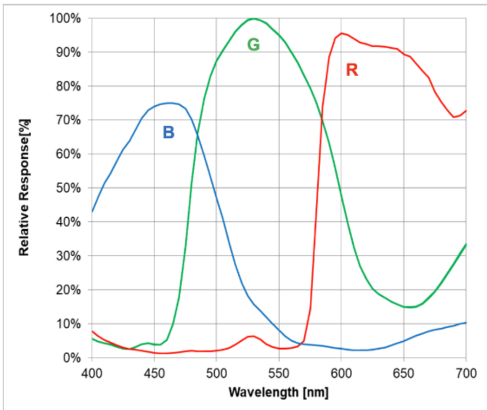


Figure 75 Spectrum-sensitivity characteristic of the Raspberry Pi camera [36]

Figure 74 also highlights another significant limitation: increasing the LED light intensity to enhance spectrum visibility can lead to interference between spectral regions—a phenomenon known as "spectral blooming" or "stray light". This effect severely compromises the accuracy of the spectrophotometer by distorting the true spectral profile.

Moreover, as shown in Figure 76, the parallel LED light source fails to provide sufficient intensity in the red region of the spectrum. This imbalance further limits the system's ability to produce accurate and complete spectral measurements, particularly in applications where red-wavelength data is critical, as shown in Figure 70 in the result section where the signal to noise ratio at red region is significantly low.

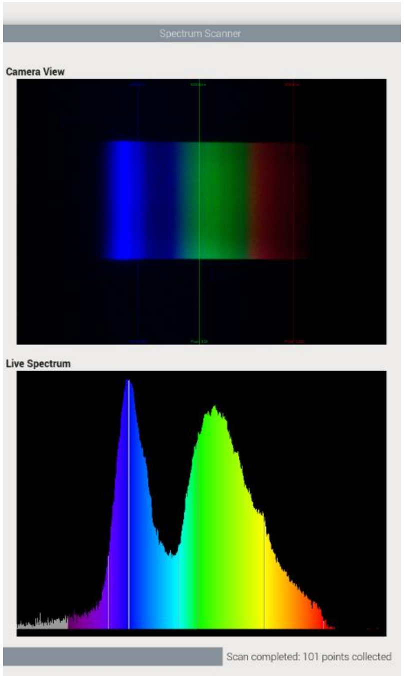


Figure 76 Parallel LED light source spectrum

Alternative broadband light sources, such as halogen bulb as shown in Figure 77, were explored to improve spectral coverage. However, these sources generate substantial heat, which poses a risk of deforming or melting nearby 3D-printed components. Additionally, halogen bulbs emit unfocused light, requiring the integration of additional optical elements—such as lenses or collimators—to concentrate the light beam for effective use in spectrometry. This added complexity introduces further design challenges in terms of thermal management, spatial constraints, and optical alignment.

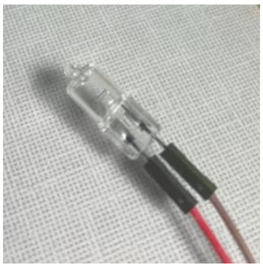


Figure 77 Halogen light bulb

Another potential solution to the limitations of a single light source is to implement a switchable light source mechanism, enabling the spectrometer to alternate between multiple light sources during sample scanning. The setup illustrated in Figure 78, incorporates a flipping mirror system that selectively reflects light from either a deuterium lamp or a halogen lamp into the spectrometer's entrance slit. This configuration allows for the combination of both light sources to achieve a broader and more continuous spectral range. The resulting composite spectrum is shown in Figure 79, demonstrating enhanced spectral coverage across both the UV and visible regions. However, the heating problem remains.

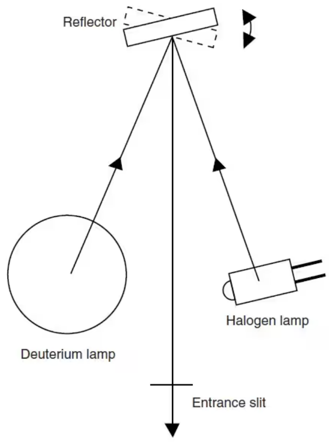


Figure 78 Illustration of Switching light source mechanism [37]

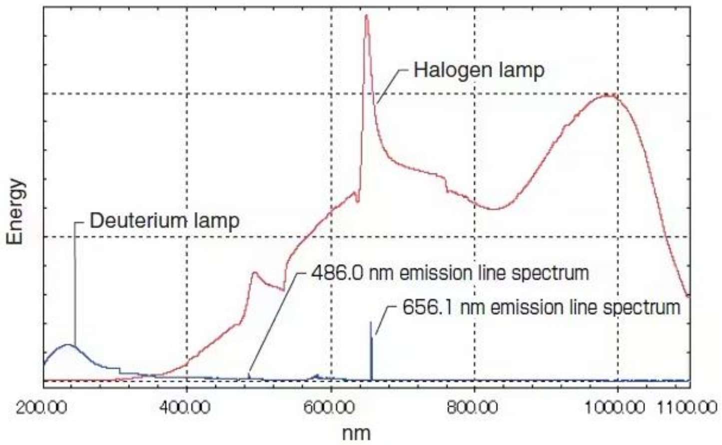


Figure 79 Combination spectrum of two light source, which achieve a better broadband cover [37]

Another limitation of the current system is its reduced sensitivity when analyzing samples with low absorption characteristics. For meaningful spectral changes to be observed, the sample must exhibit strong light absorption. This limitation is fundamentally caused by the Beer-Lambert Law, as discussed in the theory section, which relates absorbance to the molar concentration, molar absorptivity (absorption coefficient), and optical path length.

In the case of thin-layer chromatography (TLC) samples, the sample layer is extremely thin, resulting in a very short optical path length. Consequently, the total light absorption is minimal, and only compounds with both high molar concentration and high molar absorptivity produce noticeable spectral changes. For instance, during development, food dyes were tested; however, once diluted in solvent, their absorption became too weak to produce a significant difference in the spectral output.

A potential solution to the limited light absorption observed in thin-layer samples is the applying an integrating sphere into the spectrophotometer system. As depicted in Figure 80, an integrating sphere enhances light collection efficiency by diffusely reflecting incident light multiple times within a spherical cavity coated with a highly reflective material. This design allows for the measurement of total transmittance and reflectance, capturing scattered light that would otherwise be lost in conventional setups. Such a configuration is particularly beneficial for samples with low absorption or scattering properties, as it increases the effective optical path length, thereby improving measurement sensitivity.

However, incorporating an integrating sphere into a linear scanning platform presents significant challenges. The spherical geometry and size of the integrating sphere can conflict with the compact and linear design constraints of portable spectrophotometers. Additionally, aligning the optical components to ensure uniform light distribution and accurate measurements requires precise engineering.

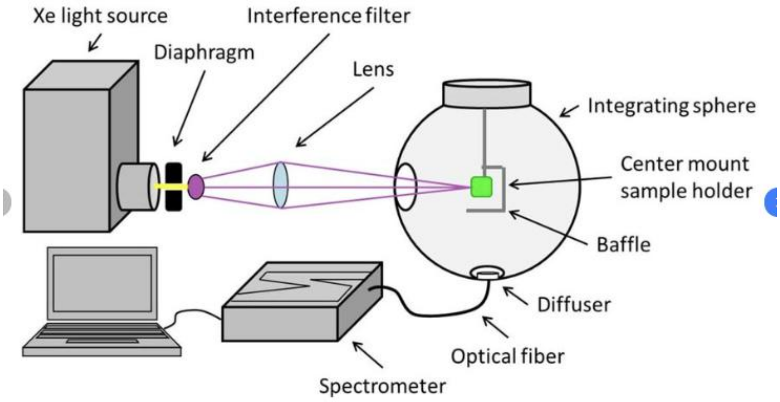


Figure 80 Illustration of how integration sphere is used in spectrometry [38]

Another promising approach to overcoming the limited optical path length in thin-layer samples is the use of multi-pass cells. As shown in Figure 81, multi-pass configurations such as the White cell allow light to traverse the sample multiple times, effectively increasing the total optical path length and enhancing the system's sensitivity. By reflecting the light beam back and forth through the sample using precisely aligned concave mirrors, the absorbance signal is amplified according to the Beer-Lambert Law, which improves detection even for low-concentration or weakly absorbing samples.

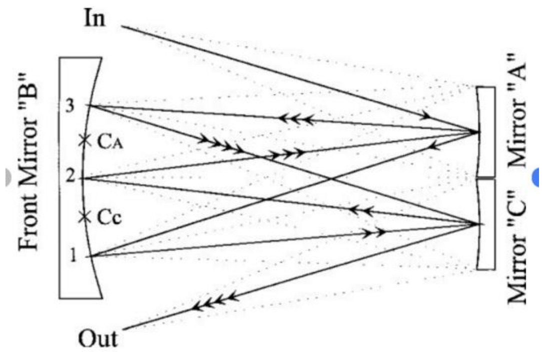


Figure 81 Side view of the classical White cell set for 8 passes [39]

However, designing appropriately scaled mirrors suitable for integration into a compact device remains a significant challenge. These mirrors must not only meet strict optical performance criteria but also be tailored to fit within the limited spatial constraints of the system. Furthermore, the mechanical vibrations generated by the stepper motor during linear movement introduce additional complexity, requiring careful compensation strategies to maintain optical alignment and measurement stability. In addition, the fabrication and precision testing of custom lenses or mirrors is both time-consuming and resource-intensive, adding to the overall development effort.

8.2 Possible improvements on linear moving platform

As highlighted in the Results section, the motor precision is currently limited due to both hardware and software factors, particularly involving the GUI. The system employs an open-loop stepper motor controlled via a TB6600 driver. In this configuration, any interference from the GUI can cause the microcontroller (MCU) to generate uneven pulse signals. This results in glitches or slips during linear motion, ultimately reducing the overall positioning accuracy.

As shown in Figure 82, a newly identified off-the-shelf closed-loop stepper motor driver presents a promising upgrade. This driver integrates a Hall-effect sensor to monitor real-time position and speed, and includes a built-in hardware PID controller for current regulation. These features enable smoother and more accurate phase control of the motor pulses. Additionally, this driver supports serial communication control or CAN control, eliminating the need for high-frequency pulse signals that depend heavily on MCU stability. Remarkably, it can even operate the motor independently with only a power supply, and it is also cost-effective, replacing the large TB6600 with only one circuit board.

The primary reason this driver was not implemented in the current system is due to its late discovery and time constraints in shipping and relative mechanical designs during the project timeline.

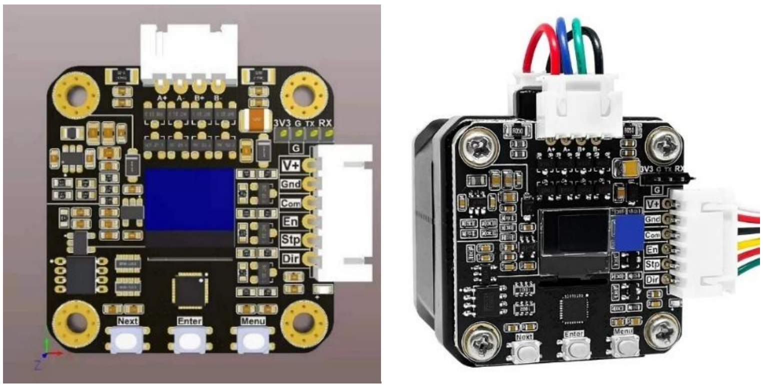


Figure 82 Circuit board of the close loop motor driver

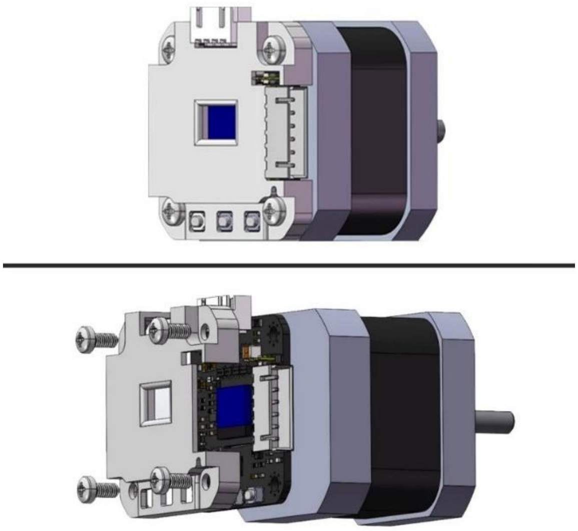


Figure 83 3D indication of how the board is fitted on the motor

Additionally, during testing some TLC samples are not moving up straight on the TLC plate as shown in Figure 84, causing the spectrometer unable to detect the sample.

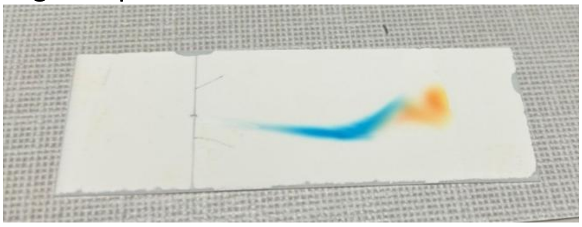


Figure 84 Off-centre samples

The possible solution is to add another axis to the platform making it able to scan in both X and Y direction.

8.3 Possible improvements on general system design

As mentioned in the Design section, the current modular design based on 3D-printed components has introduced cumulative assembly errors, which can be critically detrimental to the performance of high-precision spectrometer parts. While 3D printing offers rapid prototyping advantages, its dimensional tolerances and material limitations make it unsuitable for precision applications. An attempted improvement through SLA (stereolithography) printing yielded components with finer surface finishes; however, excess resin residue accumulated in tight corners as shown in Figure 85, and the parts were prone to cracking when attempting to use heat-set inserts as shown in Figure 86, which are incompatible with the brittle nature of cured SLA resin.

Figure 85 Corner excess resin residue

Figure 86 Cracks caused by heat inserts

To address these limitations, an alternative solution is the use of CNC-machined aluminum components. These offers significantly improved dimensional accuracy, structural rigidity,

and allow for reliable threaded connections without requiring heat inserts. This makes them more suitable for precision alignment and repeatable assembly.
While the current 3D-printed parts suffice for rapid development, their structural integrity remains a concern—especially under impact or load—due to the inherent fragility of common FDM materials. As a potential compromise, using stronger materials such as glass fiber or carbon fiber plates could enhance durability. These materials can be easily fabricated through online manufacturers. However, their longer lead times for shipping and processing make them less ideal for rapid prototyping, same as CNC, especially when compared to the single-day turnaround of in-house 3D printing.
As mentioned in the Design section, the current power delivery setup relies on a built-in power strip to supply all components within the device. While this approach offers convenience during prototyping, it limits the system's portability, and may introduce electrical noise between modules.
A more robust and scalable solution involves transitioning to a battery-powered system using voltage divider and voltage regulator circuits, as illustrated in Figure 87 and Figure 88. These circuits would enable precise and stable power distribution to various components—such as the microcontroller, sensors, spectrometer, and motor drivers—by converting battery voltage to the specific levels each module requires (e.g., 5V, 12V). This approach also supports off-grid, mobile usage, and contributes to cleaner power delivery and a more modular, embedded design.
However, implementing this solution would also require consideration of USB-powered components within the system. This would necessitate a complete PCB redesign to integrate new power management components and USB port routing, along with custom PCB fabrication and manual soldering of components.
Due to time constraints and the late-stage discovery of this improvement, the power optimization solution—though promising—was not realized in the current iteration of the device.

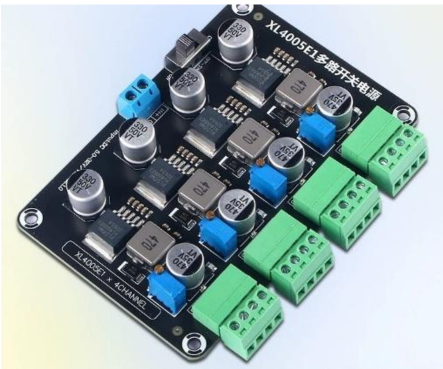


Figure 87 Voltage divider circuit

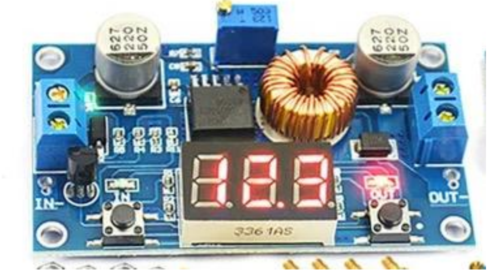


Figure 88 DC-DC Voltage regulator

In consider of battery, as shown in chart 2 in the design section, the overall power consumption for all components is about 63.4W, take a 12V 11200mAh battery shown in Figure 89 as an example:

Figure 89 12v 11200mAh battery

The operation time could be calculated as follows:

$$Time\ of\ operation(h) = \frac{11.2Ah\ \times 12V}{63.4W} \approx 2.12h$$

Which is already enough for 250+ tests if assuming each test is 30s as current performance.

Additionally, about the thread conflict between motor control and GUI activity, due to limited time and experience with concurrency programming in Python, the exact root cause of the thread conflicts remains uncertain.

One likely contributing factor is the inherent limitation of Python's Global Interpreter Lock (GIL), which restricts true parallel execution of threads within a single process. Python's built-in threading module is often suitable for I/O-bound tasks, but not ideal for CPU-bound operations or scenarios requiring tightly timed parallel control, such as motor pulse generation or real-time GUI updates.

A potential alternative is to implement multi-processing, which spawns separate processes that run independently, bypassing the GIL and enabling true parallel execution. This approach may offer better performance and thread isolation. However, it also introduces additional complexity in terms of inter-process communication, data synchronization, and debugging.

9. Reflection on learning

Throughout the development of this project, I acquired a broad spectrum of technical skills that deepened my understanding of interdisciplinary system integration, from mechanical hardware to software architecture.

Spectrometry Data Processing:

Engaging with spectral data required not only a solid grasp of optical theory but also practical implementation of data extraction and calibration techniques. By incorporating dynamic RGB channel weighting and polynomial wavelength calibration, I learned how to process raw image data from the camera into accurate spectral plots. This sharpened my skills in signal analysis, curve fitting, and data visualization—particularly for applications where measurement fidelity is critical.

PyQt Design:

Designing the user interface with PyQt5 significantly advanced my experience in GUI development. I learned to structure a complex, multi-threaded interface that not only displayed real-time spectral data but also offered interactive calibration and scanning controls. The signal-slot architecture of PyQt was particularly instrumental in maintaining a responsive user experience, even while handling concurrent hardware and data processing tasks.

Raspberry Pi Application:

The Raspberry Pi 4B served as the core processing unit of the system, allowing me to delve into Linux-based development, GPIO handling, and interfacing with both sensors and actuators. Working with this platform taught me to optimize software performance under hardware constraints and integrate various I/O protocols reliably in a real-time application environment.

Stepper Motor Control:

Implementing precise motion control using a stepper motor and a TB6600 driver introduced me to open-loop motion systems and their limitations. I developed threaded motor control routines that ensured safe, accurate positioning through endstop detection and signal-based motion management. This experience expanded my knowledge of motion control principles and embedded system timing.

Multithreading:

Handling real-time scanning, spectrum acquisition, and GUI updates required robust multithreaded programming. This challenged me to manage thread safety, synchronization, and system responsiveness. I became familiar with Python's threading limitations (e.g., the Global Interpreter Lock) and began exploring multiprocessing as a more scalable alternative.

SolidWorks Design:

All custom mechanical components were designed in SolidWorks, enabling me to refine my skills in parametric modeling and mechanical tolerance analysis. The design process emphasized practical considerations like load paths, modularity, and component fit—especially important when integrating 3D printed parts with electronics and optics.

3D Printing Experience:

FDM 3D printing was essential for rapid prototyping and iteration. Through repeated cycles of design, slicing, and fabrication, I learned how part orientation, material choice, and print settings influence structural performance and dimensional accuracy. Additionally, I identified the trade-offs between printability and precision, which informed decisions about modularity and assembly.

10. Conclusions

This project successfully developed a compact, low-cost, and modular robotic scanning platform integrated with an optical spectrophotometer for chemical imaging. Using a Raspberry Pi 4B, stepper-motor-driven linear platform, and a PyQt5-based interface, the system achieved real-time spectrum acquisition, visualization, and storage. Tests confirmed accurate spectral detection, reliable scanning, and the ability to distinguish subtle variations in samples such as solid color plates and thin-layer chromatography slides.

The design prioritized modularity and portability, though limitations were noted, including platform motion accuracy affected by GUI processes, light source spectral gaps, and reduced sensitivity for low-absorbance samples. Potential improvements include closed-loop motor control, upgraded broadband light sources, and CNC-machined precision parts.

Overall, the project demonstrates strong potential for real-world applications in remote healthcare, emergency diagnostics, and scientific research. It also provided valuable interdisciplinary learning, integrating mechanical design, embedded systems, optical engineering, and software development into a functional, user-friendly platform.

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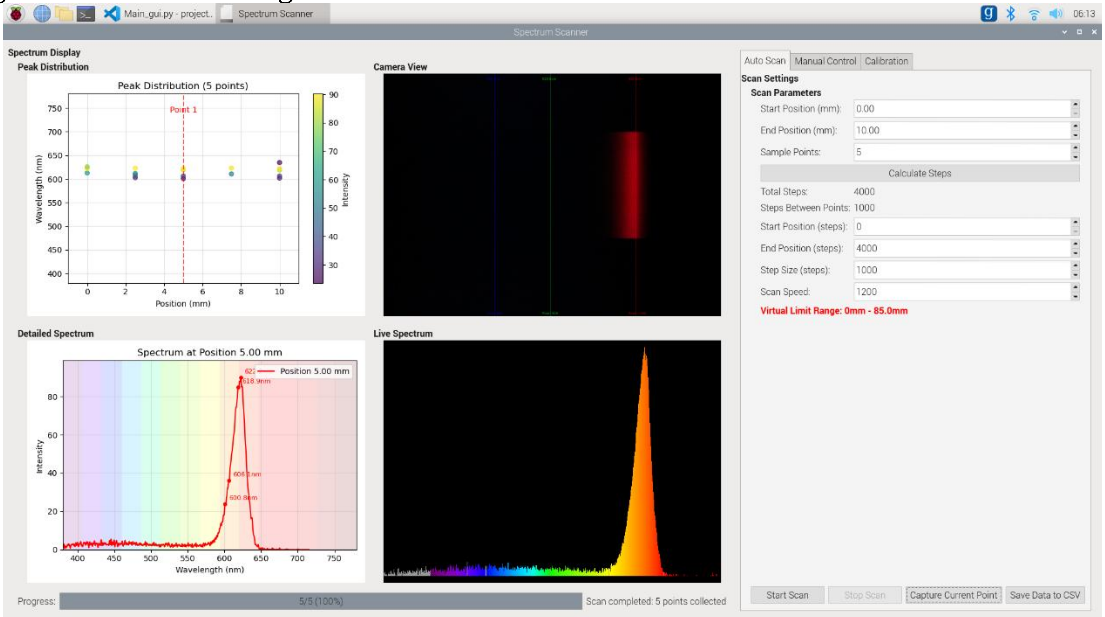
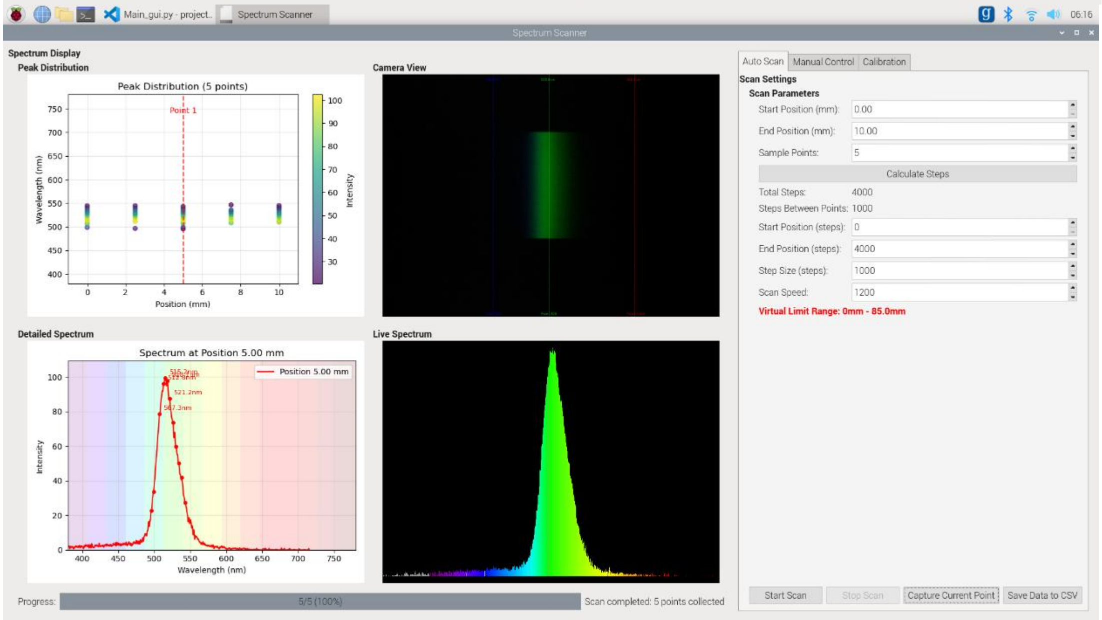
Appendix

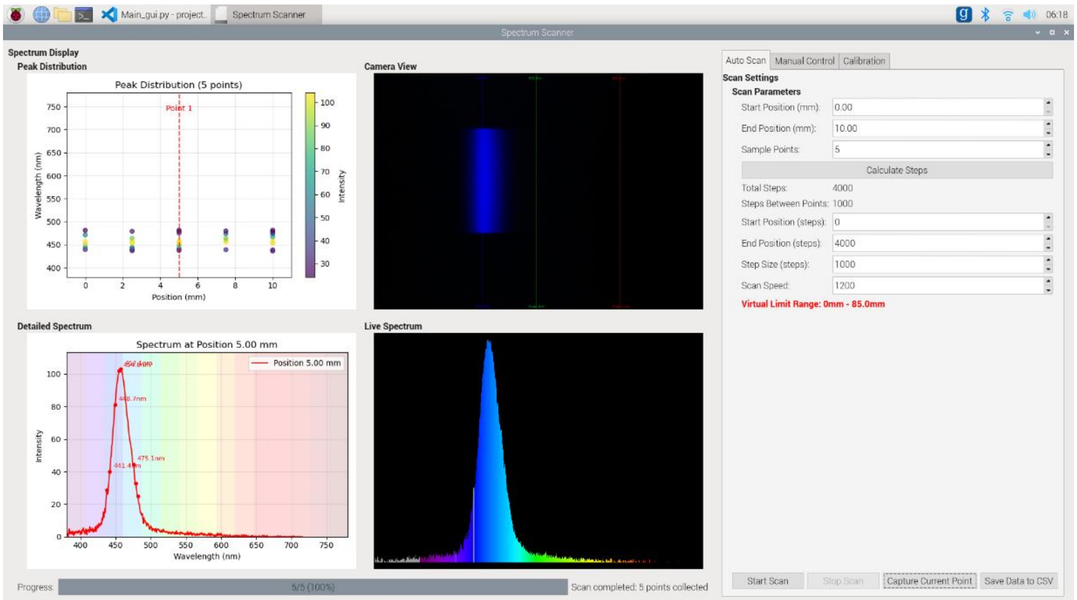
Appendix 1

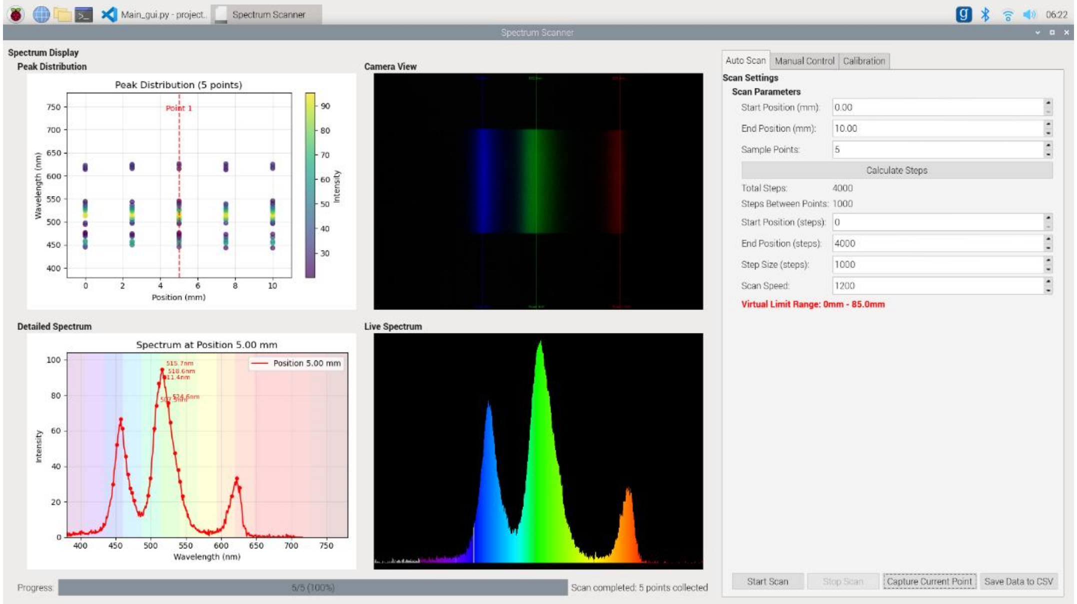
Appendix 2

original screenshot for Figure 65



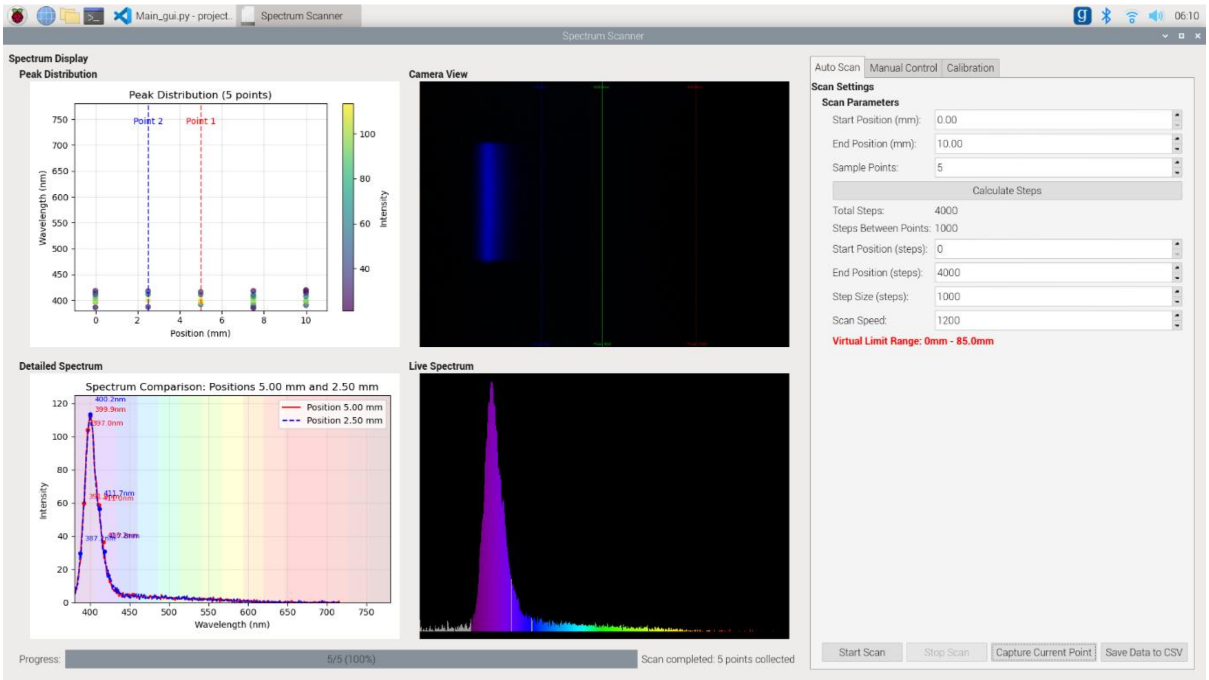






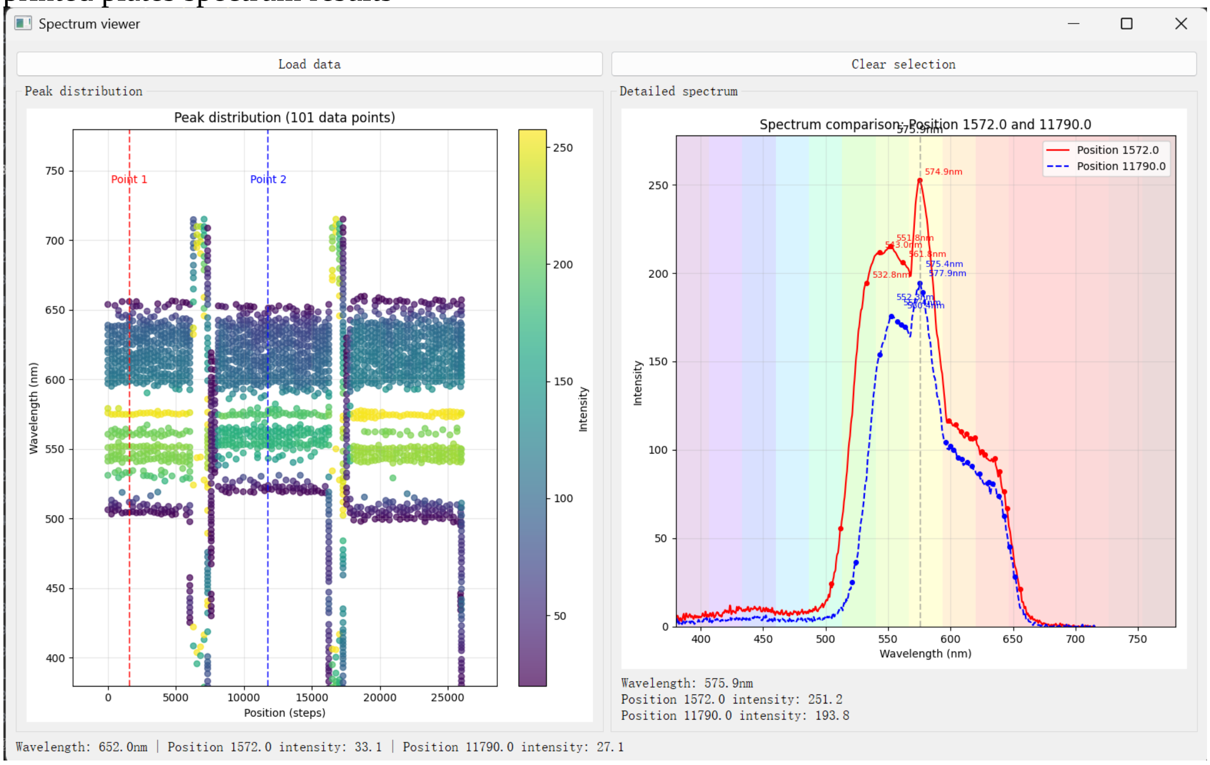
Appendix 3

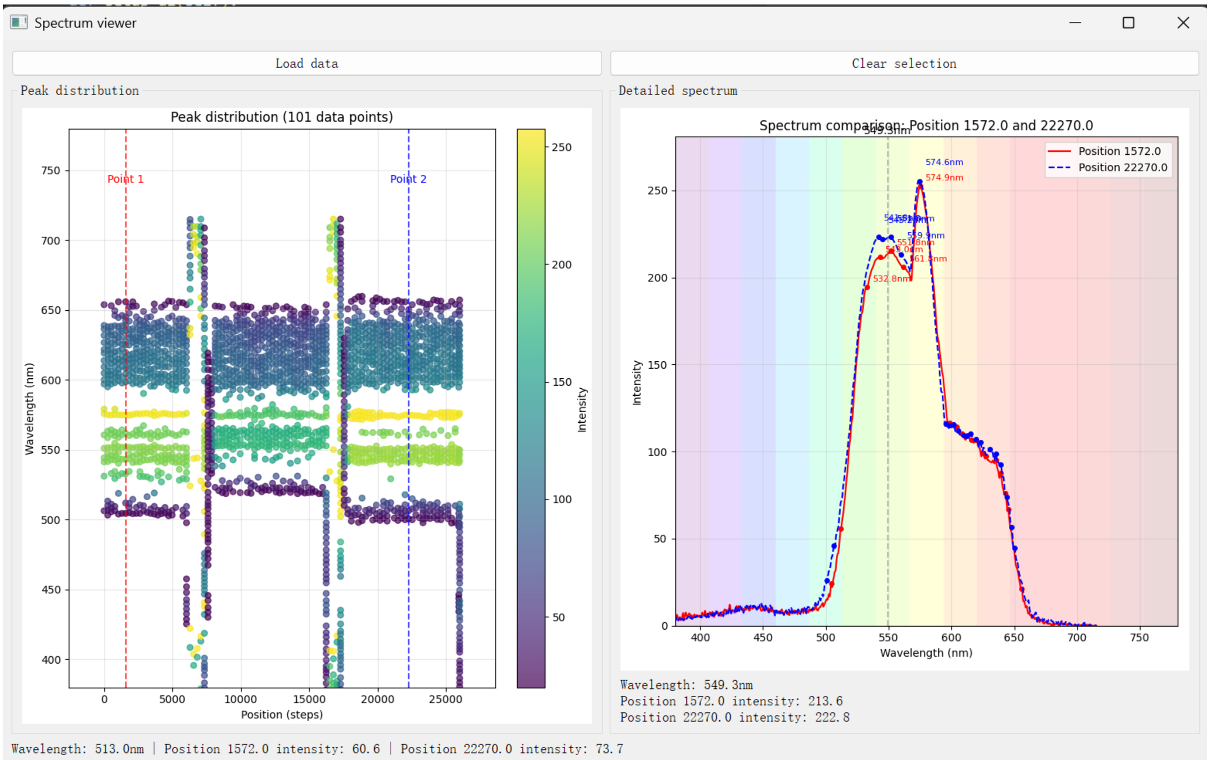
UV resulting spectrum



Appendix 4

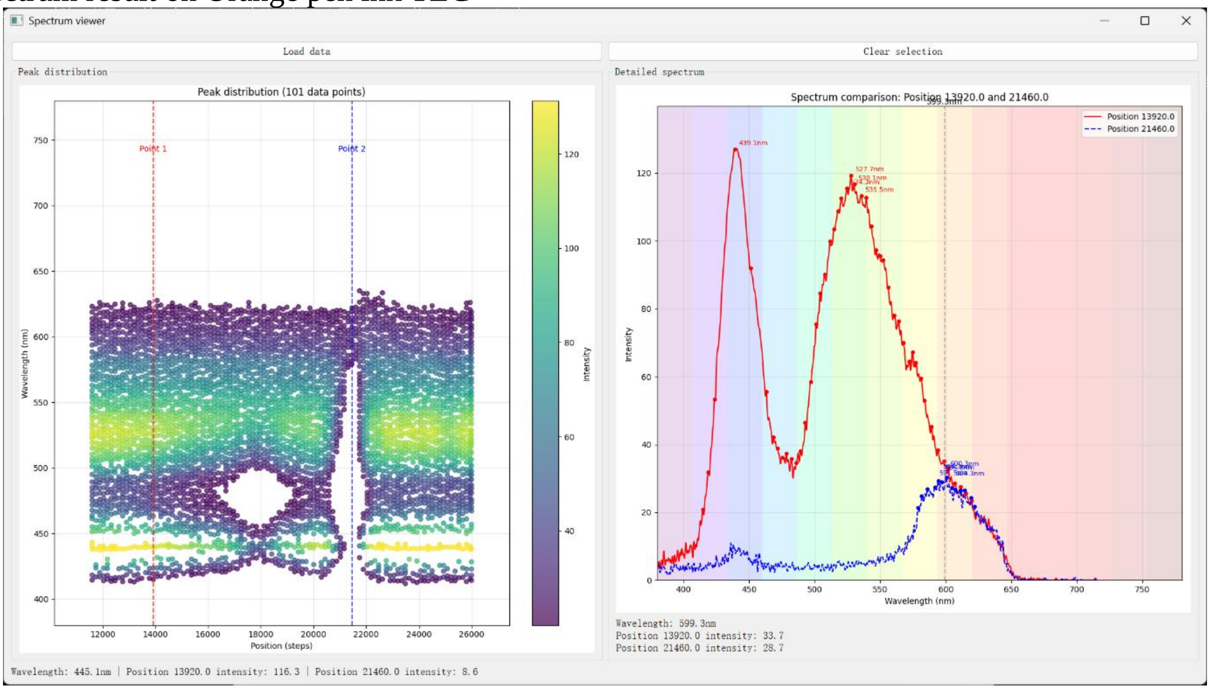
3D printed plates spectrum results

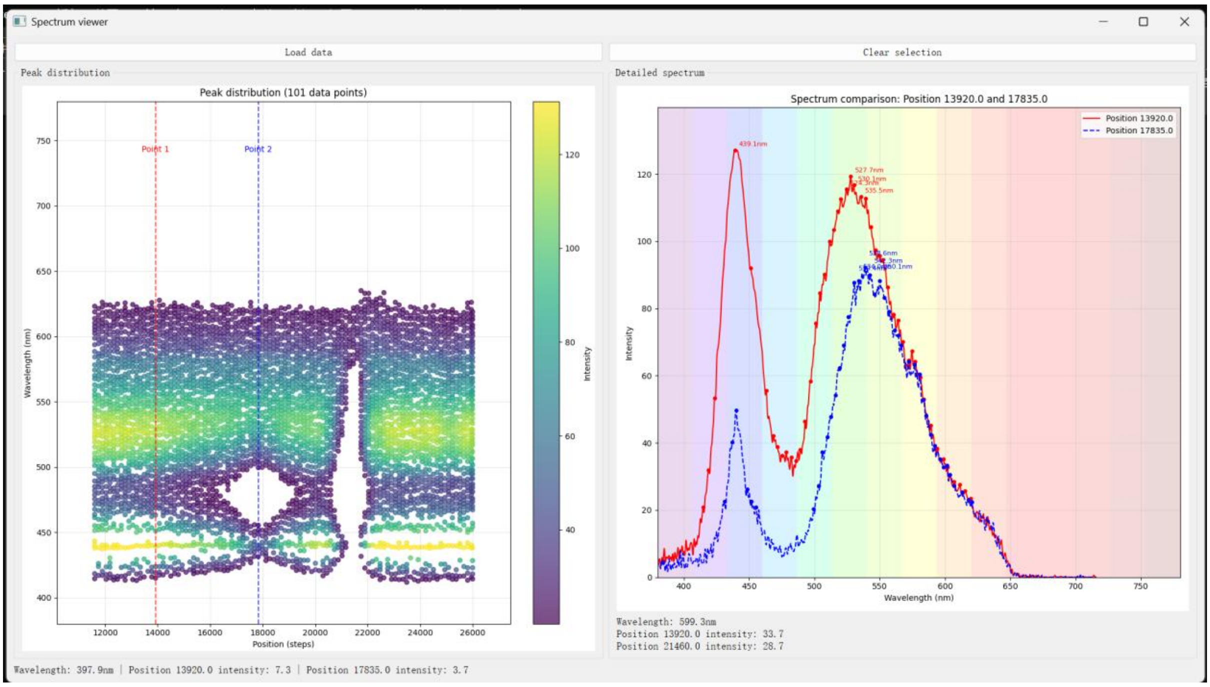




Appendix 5

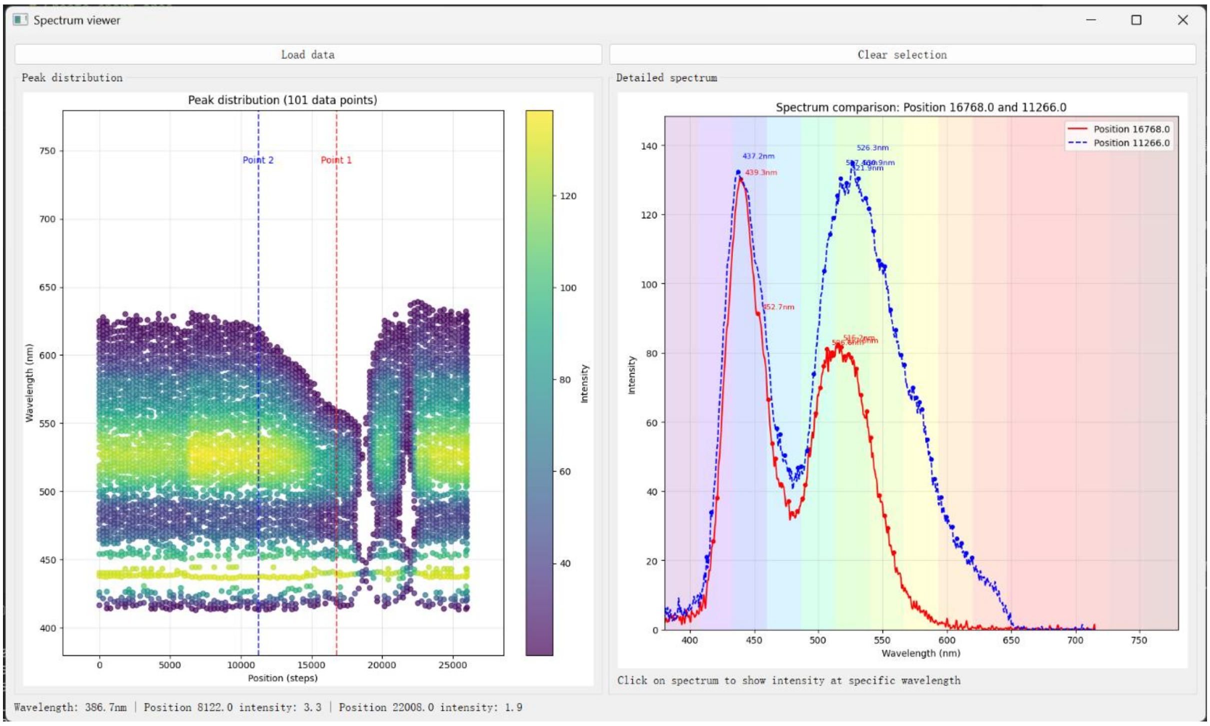
Spectrum result on Orange pen ink TLC





Appendix 6

Spectrum result of Red and Blue mix TLC



Appendix 7

Key specifications

Parameter	Target	Verification
Linear Movement Accuracy	±0.5mm over 100mm	Can be verified using a Vernier caliper or calibrated laser displacement sensor
Spectrometer Wavelength Range	400nm to 700nm	Test against known reference light sources
Spectrometer Resolution	1nm	Compare detected spectra with reference materials
Total process time	15-20s	Measure while demonstrating the process
User Interface Delay	Less than 1ms	Able to respond to interrupt control instantly

Appendix 8

Gantt chart

work-packages	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	===
Work package 1																
Task 1: Identified and documented components for each system section.																
Task 2: Searched for suitable components online.																
Task 3: Explored relevant GitHub repositories and project videos.																
Task 4: Researched academic papers on improving linear motion and spectrometer performance, focusing on chemical analysis.																
Work package 2																
Task 1: Test Ubuntu 20.04 and Raspbian on separate SD cards, running Python and C++ codes.																
Task 2: Follow YouTube tutorials and forums like StackOverflow for efficient environment setup.																
Task 3: Connect and test a portable screen for data and UI display functionality.																
Work package 3																
Task 1: Connect motor, power supply, and driver to Raspberry Pi based on seller's instructions.																
Task 2: Send step signals from MCU to motor to test movement.																
Task 3: Measure platform movement per motor rotation to ensure accurate, repeatable control.																
Work package 4																
Task 1: Design 3D models for the platform and supports in Solidworks, ensuring precise alignment for the spectroscope and camera.																
Task 2: Use a personal 3D printer for rapid prototyping and iteration of components.																
Task 3: Calibrate the spectrometer using known light sources for accurate spectral detection and analysis.																
Work package 5																
Task 1: Modularized specific functions for easy integration with the main display interface.																
Task 2: Designed the user interface layout and arranged all functionalities.																
Task 3: Enhanced overall performance through optimized storage management.																

Work package 6																
Task 1: Implement PCA algorithm to process spectrum data and identify chemical compositions.																
Work package 7																
Task 1: Use Solidworks to model the structure, ensuring secure component housing with proper spacing.																
Task 2: Incorporate shielding to block external light for optimal device operation.																
Task 3: Add features to improve robustness and give the design a more product-like quality.																
Additional adjustments																

Appendix 9

Illustration of final device